Getting the Most out of HEX

Mohamed, Manuel & Elisabeth
July 10, 2025

Public

Agenda

1 Introduction

2 Calculation Engine

3 HEX Engine

4 Troubleshooting

Introduction

Introduction

Hana EXecution Engine or HEX for short is HANA's unique execution engine for all relational queries

Replaces legacy engines

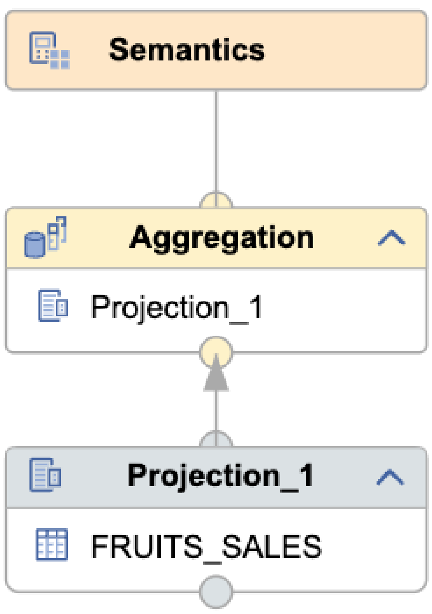
State-of-the-art features and technologies

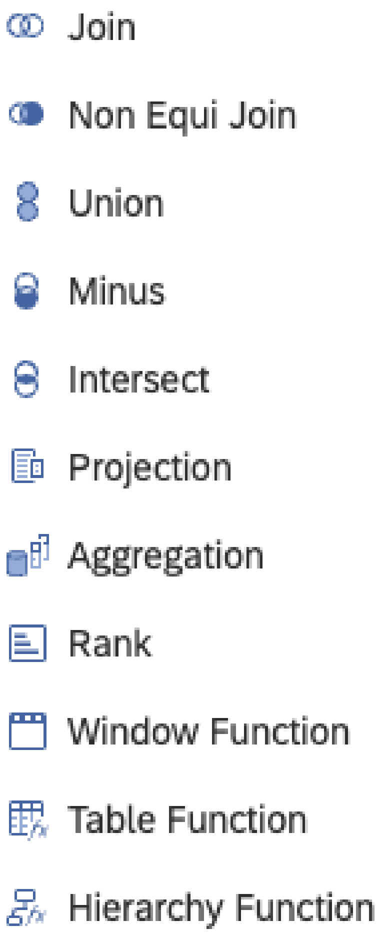
Improved perfomances

Calculation Engine

Calculation Engine Introduction

- Execute Calculation Views
- CalcViews: graphically modeled queries
- BAS: modeling tool for HC
- BAS offers all relational operations as calculation nodes
- CE supports other non-relational operations used internally: MP, AggrWithHierJoin, customCpp etc.
- Advantages:
  - no-code (SQL) approach
  - reusability: CalcViews are catalog objects, can be used in any SQL query or as datasource in other CalcViews

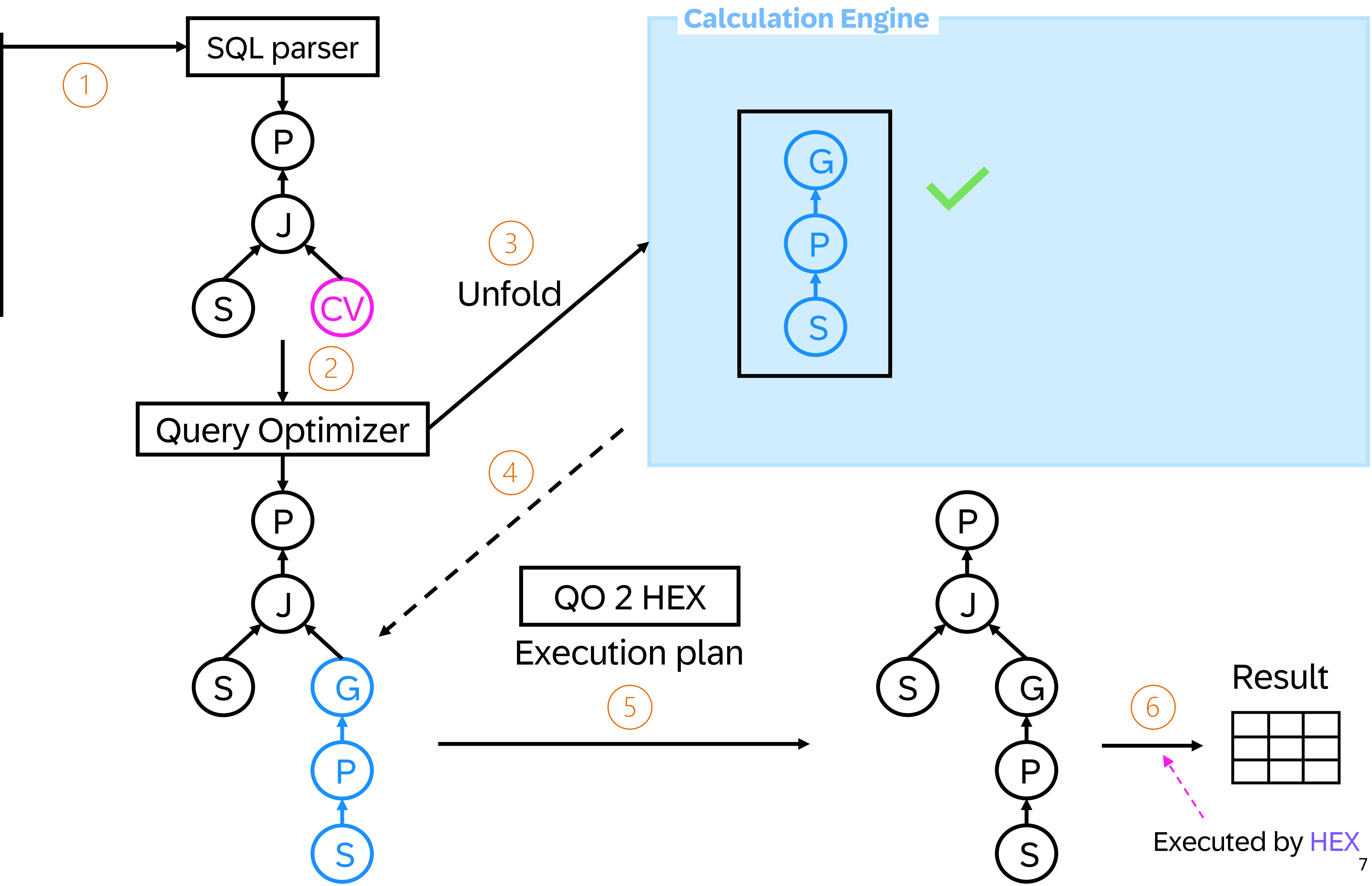




Unfolding: Simplified Workflow

QUERY

```
SELECT T.A, CV.B
FROM table as T
JOIN CalcView AS CV
ON T.id = CV.id
WHERE T.B = x
```

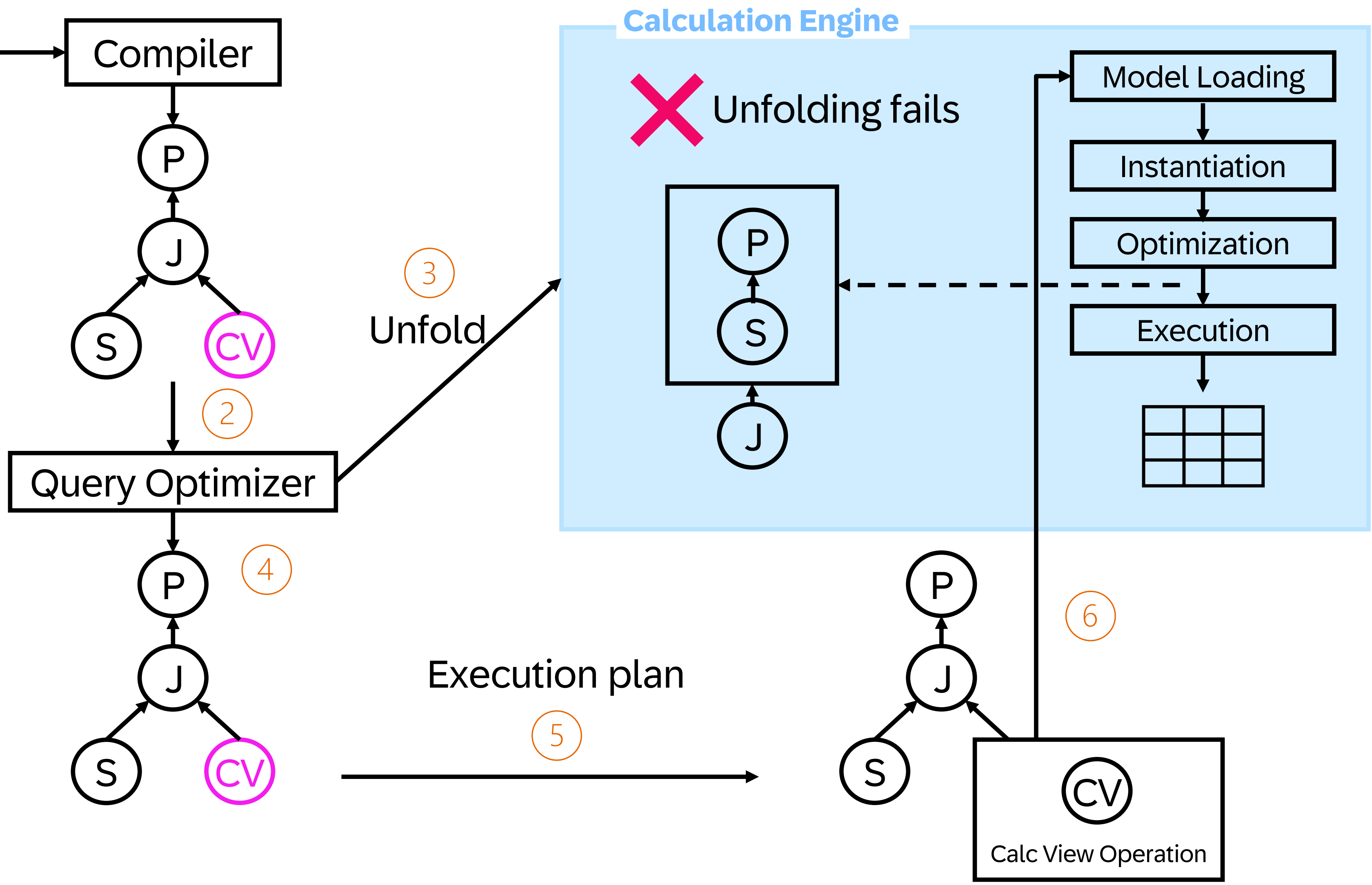


P	Projection
J	Join
S	Selection
CV	Calculation View
G	Group By

Partial Unfolding: Simplified Workflow

QUERY

```sql
SELECT T.A, CV.B
FROM table as T
JOIN CalcView AS CV
ON T.id = CV.id
WHERE T.B = x
```



P	Projection
J	Join
S	Selection
CV	Calculation View
G	Group By

Calculation Engine Tracers

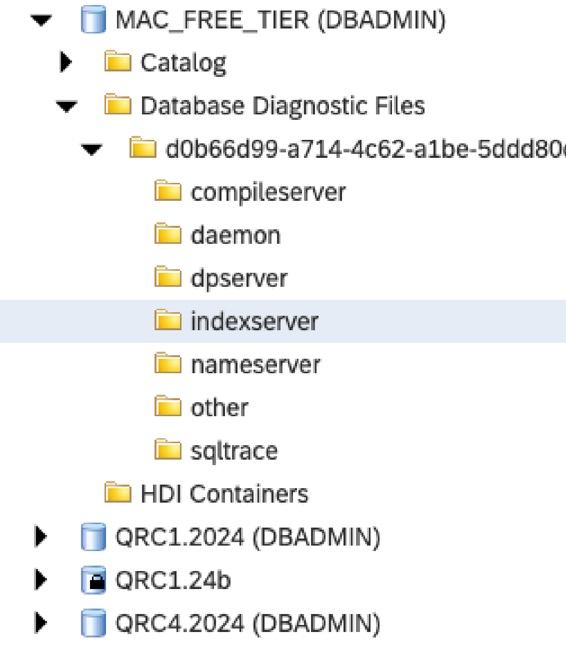
- Calcengine: most generic tracer,
    - contains: query (search object), scenario loading, scenario cache hit/miss, authorization, AP checks ,

- ceInstantiate: instantiation traces (inc. final instantiated model)
- ceOptimizer, ceOptimizerInfo: optimization traces, rules applied, intermediate runtime models, final optimized Runtime model

- Ceqo, ceplanexec, ceexecute: execution traces. QO plan generation, POPs generation, QOPlan cache for QoPops

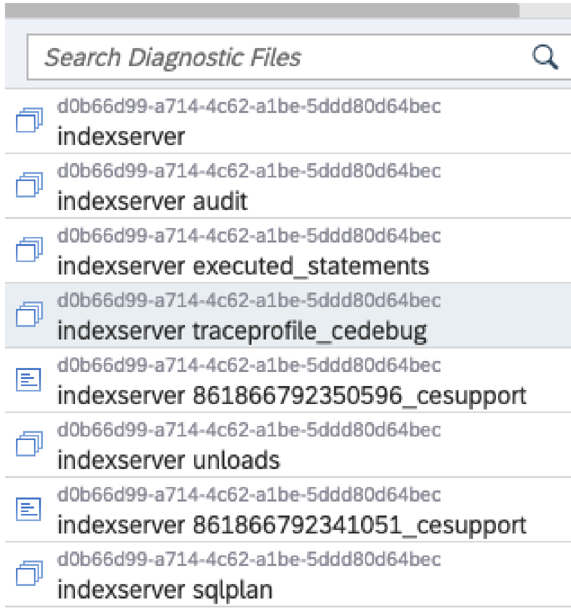
How to Generate ce Debug Traces

1. Using tracing profile



Trace file name   tracer   Level   Exclude user   Restrict to user

```
alter system alter configuration ('global.ini','SYSTEM') SET ('traceprofile_cedebug', 'sql_user') = '!_SYS_STATISTICS' with reconfigure;
alter system alter configuration ('global.ini','SYSTEM') SET ('traceprofile_cedebug', 'sql_user') = 'DBADMIN' with reconfigure;
ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM' ) SET
('traceprofile_cedebug', 'calcengine') = 'debug' ,
('traceprofile_cedebug', 'ceoptimizer') = 'debug' ,
('traceprofile_cedebug', 'ceoptimizerinfo') = 'info' ,
('traceprofile_cedebug', 'ceinstantiate') = 'debug',
('traceprofile_cedebug', 'ceplanexec') = 'debug',
('traceprofile_cedebug', 'ceqo') = 'debug',
('traceprofile_cedebug', 'ceexecute') = 'debug',
('traceprofile_cedebug', 'maxfilesize') = '200000000'
WITH RECONFIGURE;
```



How to Generate ce Debug Traces

2. Using variable $$CE_SUPPORT$$

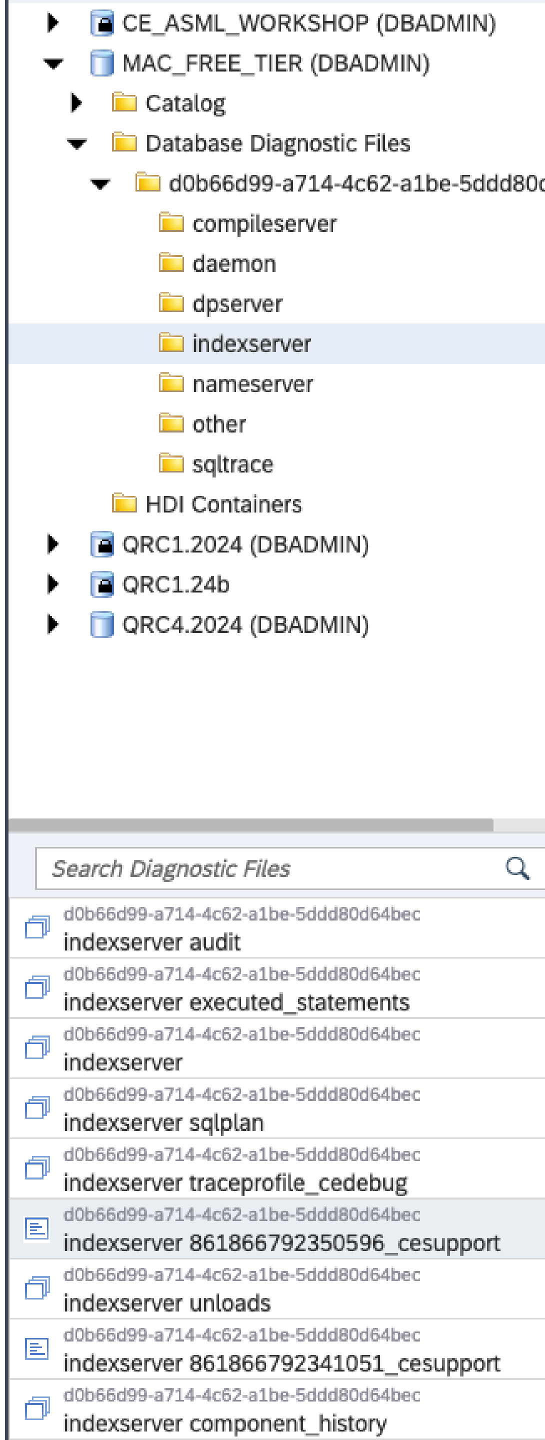
This works only with CE tracers

```
SELECT E.NAME, SUM(S.AMOUNT) AS REVENUE
FROM "CE_ASML_DEMO"."SALES" AS S
INNER JOIN "CE_ASML_DEMO"."SIMPLE_PROJ" AS E
ON S.EMP_ID = E.ID
GROUP BY E.NAME
WITH PARAMETERS('PLACEHOLDER' = ('$$CE_SUPPORT$$', 'd_calcengine, d_ceinstantiate,
d_ceoptimizer, i_ceoptimizerinfo, d_ceqo, d_ceplanexec, d_ceexecute'))
```

l_tracer

$$l \in \{e, w, i, d\}$$

e = error, w=warning, i=info, d=debug



How to Read ce Traces

Trace file structure in case of full unfolding

The search object (query)

Stored models loading (AP checks, cache check)

Runtime model instantiation

Instantiated model comes after: **Instantiated model**

Runtime model optimization

optimized model comes after: **Optimized model**

```
[2147]{200669}[1066/-1]<192d774a7295037f37b12f23c17cb938#0000000000000000> 2024-10-29 09:30:12.477310 d ceQo          ceController2.cpp(01555) : Generated QO Tree:
# PROJECT (opId:L:9) (engId:L:9) ((3, 1), (3, 0)) result_size:-1 rel_addr:0x00007f710bc0c950
  # GROUP BY (4) (opId:L:8) (engId:L:8) COLUMNS:: (3, 1), (3, 0) input_size:-1 result_size:-1 rel_addr:0x00007f710bc0c720
    # TABLE EMPLOYEE (3) (opId:L:7) (engId:L:7)  TABLE used cols::  TABLE histo cols::  TABLE key joined cols::  input_size:-1 table_location: NULL|65535 result_size:-1
        rel_addr:0x00007f710bc0bde0
[2147]{200669}[1066/-1]<192d774a7295037f37b12f23c17cb938#0000000000000000> 2024-10-29 09:30:12.477314 d CalcEngine    ceController2.cpp(01566) : View 'CE_ASML_DEMO
    :SIMPLE_PROJ' successfully unfolded
```

Ce_debug_trace

Summary

What did we see:

- How to generate and read calcengine debug traces
- How to check unfolding/partial unfolding
- Difference between unfolding/partial unfolding
- Impact on performance of partial unfolding

HEX Engine

HANA Execution Engine

Why HEX?

HEX in a Nutshell

Recommendations

Why HEX?

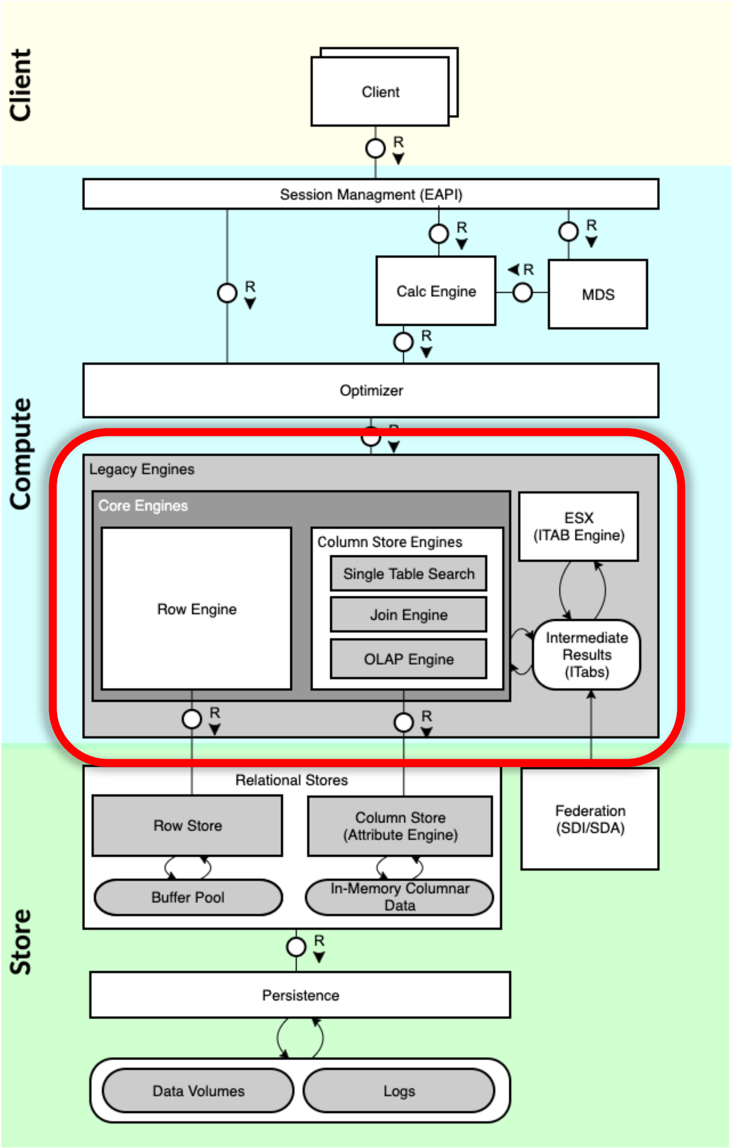
Where are we coming from?

A plethora of Execution Engines…
… focusing on single aspects of Query Execution.

- Column Store
  - **J**oin **E**ngine: Join Queries, based on Semi-Join Reduction, Specialized for N-way Joins
  - **O**LAP **E**ngine: Star Schema, Map-Reduce—like Processing Model
  - **S**ingle **T**able **S**earch: Simple `SELECT` Queries
- Row Store
  - ESX: Row Store

Limitations

- Materialization at relation boundaries
- Difficulty in global optimization using heterogenous engines



HEX – One Engine

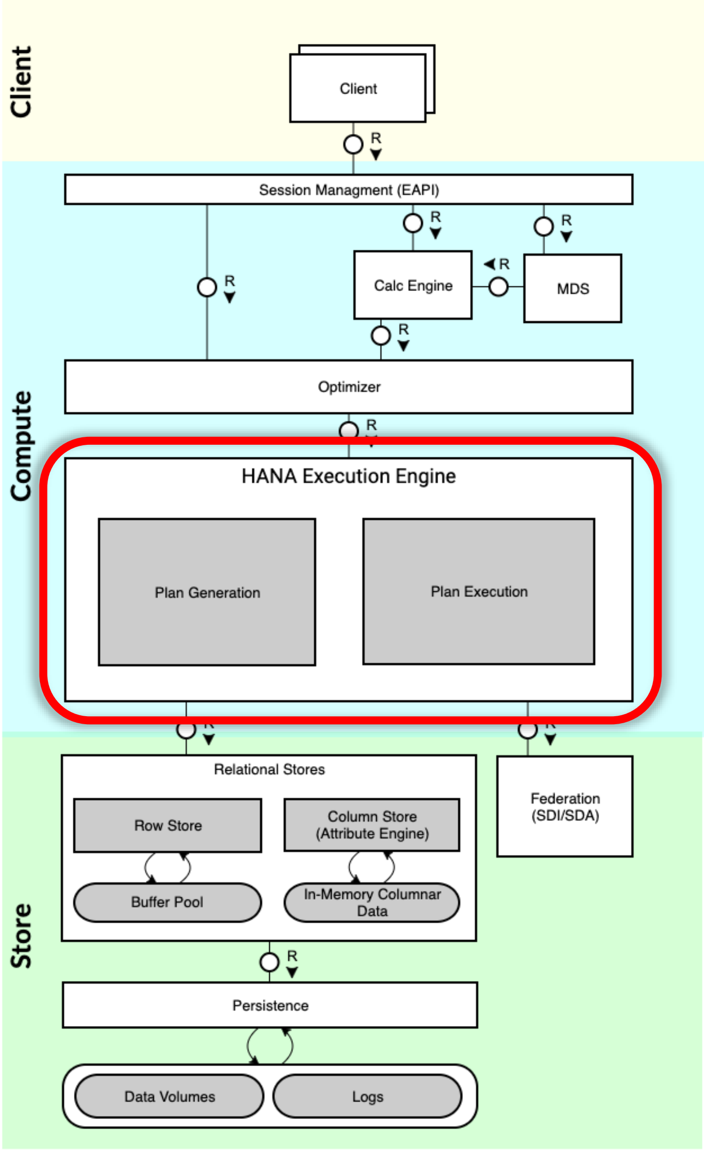
State of the Art

- **Pipelining:** Reduce memory consumption, improve cache behaviour
- **Code Generation:** Pragmatic, not brute-force but use it where it makes sense
- **Result streaming:** Produce result rows as soon as possible
- **Data Driven Parallelism:** based on the structure and volume of the actual data

Modular & Flexible

- extension basically means adding new operators
- no monolithic design assumptions like in JE or OE
  → fully composable engine

Natural integration with SQL optimizer

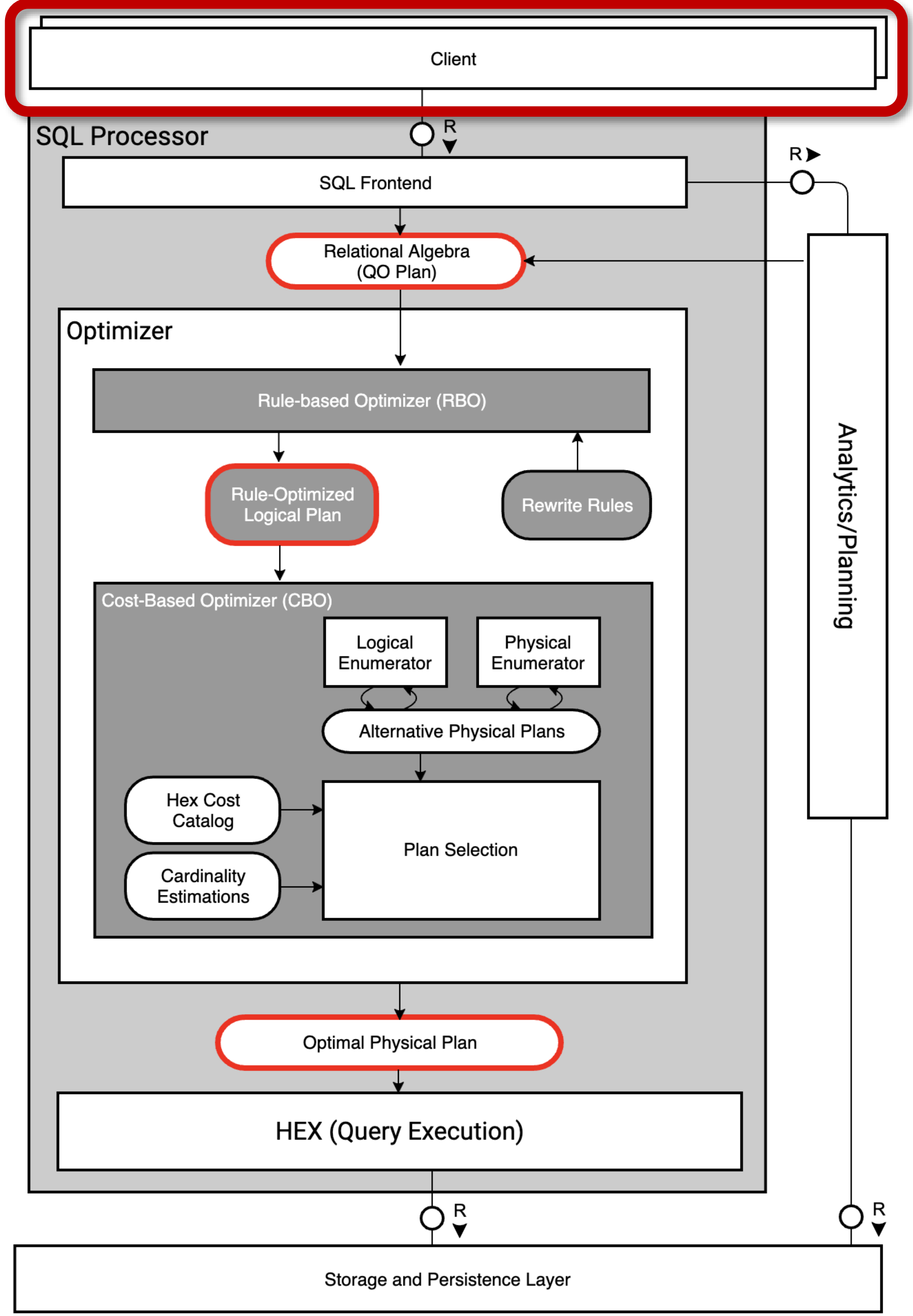


HEX in a Nutshell

Client

Our journey starts with a SQL query

```sql
SELECT R.id, S.date
  FROM R, S
 WHERE R.id = S.id
   AND S.val > 10
```

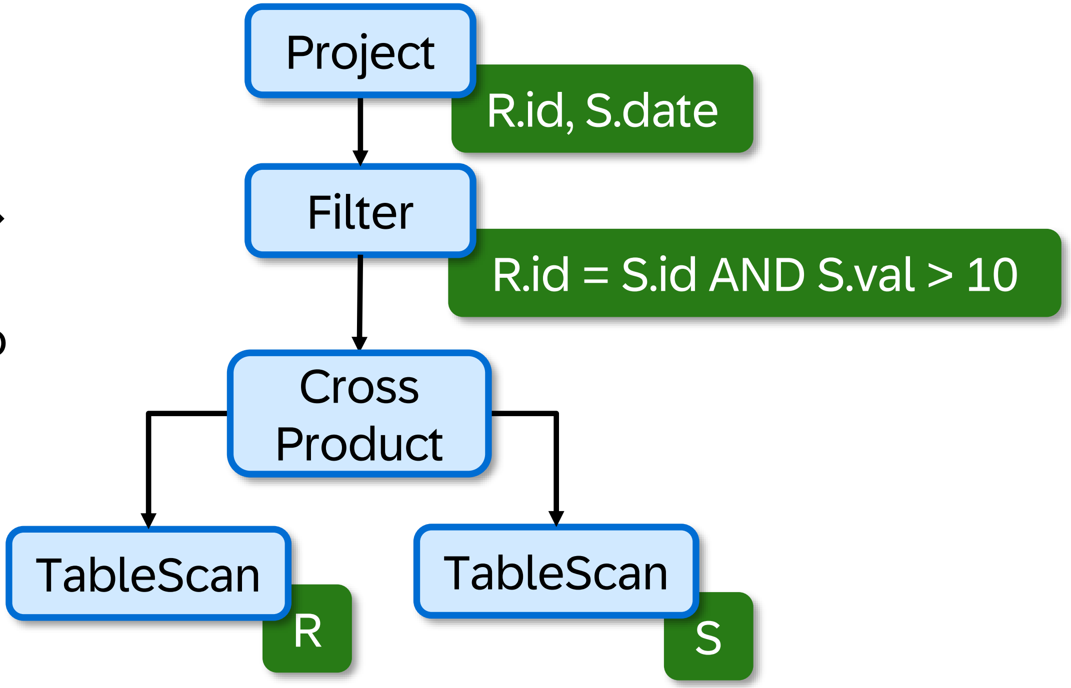


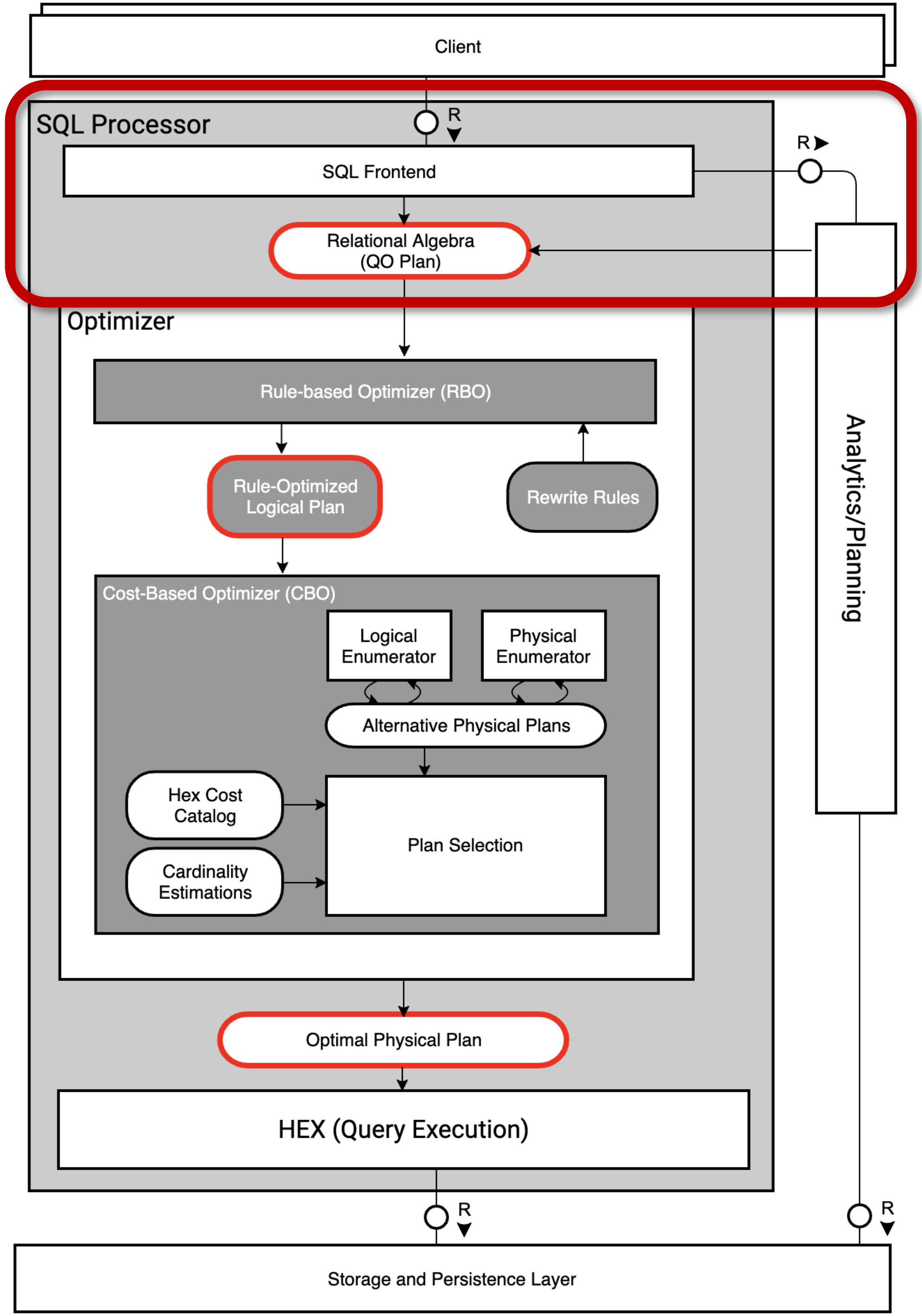
SQL Frontend

Our journey starts with a SQL query

```
SELECT R.id, S.date
  FROM R, S
 WHERE R.id = S.id
   AND S.val > 10
```

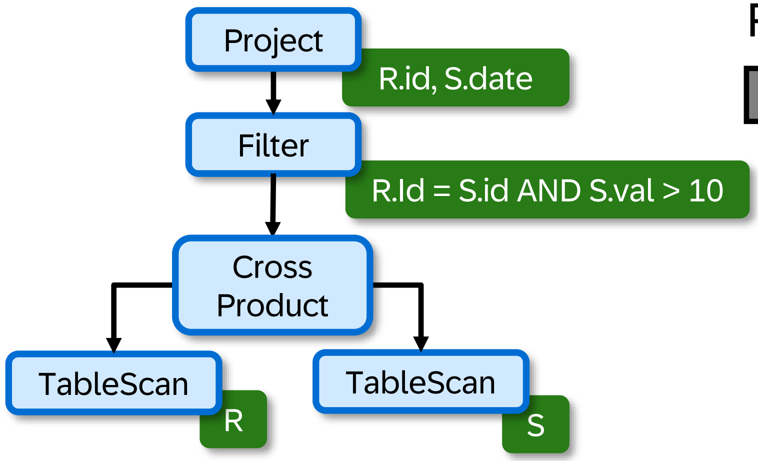
Translation into Canonical Plan



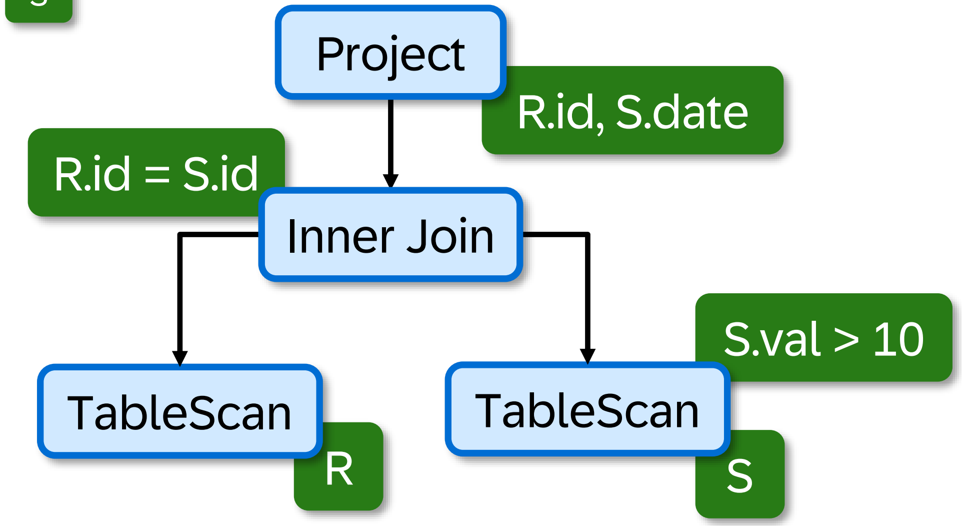


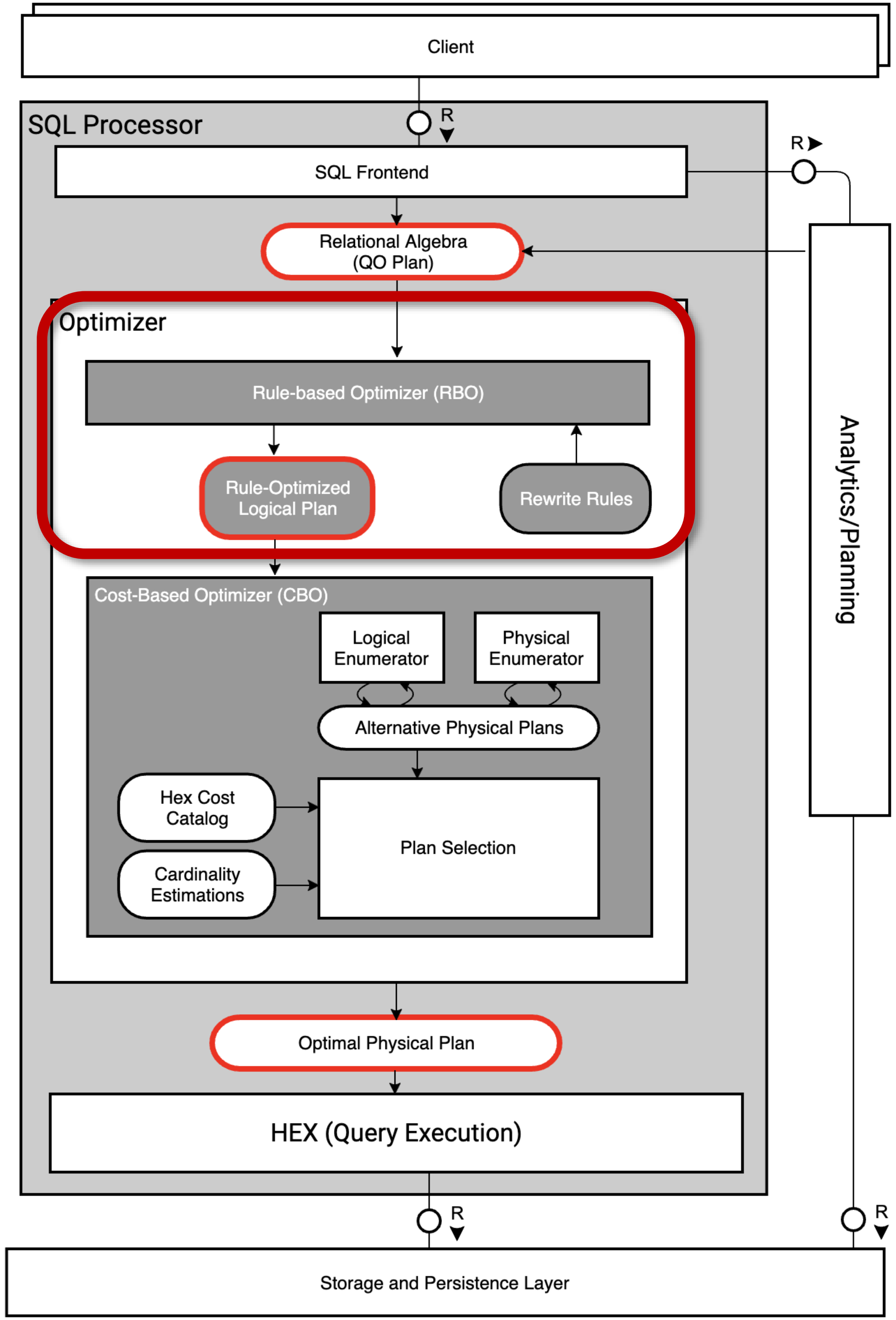
Optimizer (Rule-based)

Canonical Plan



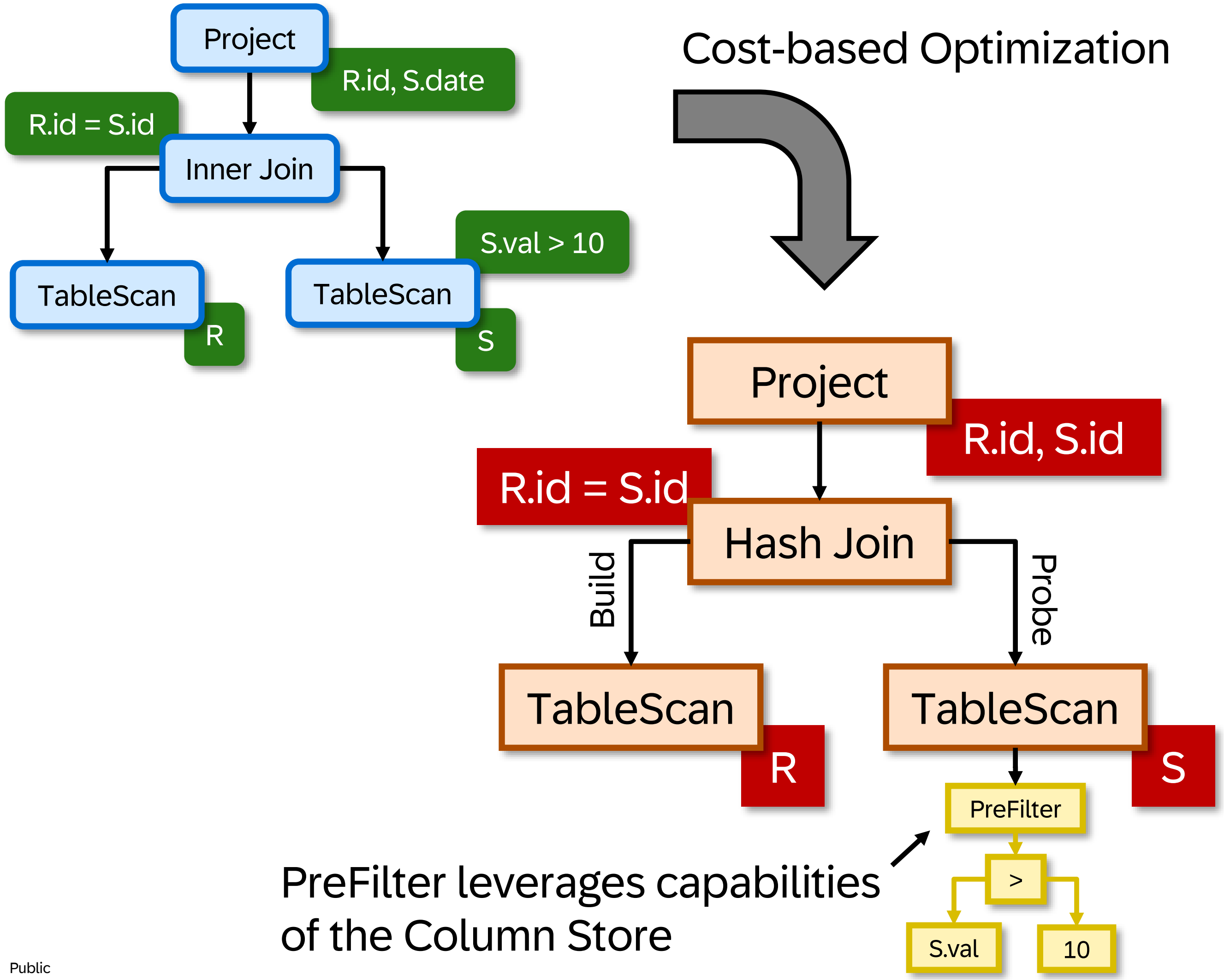
Rule-based Optimization
(i.e Predicate Pushdown)

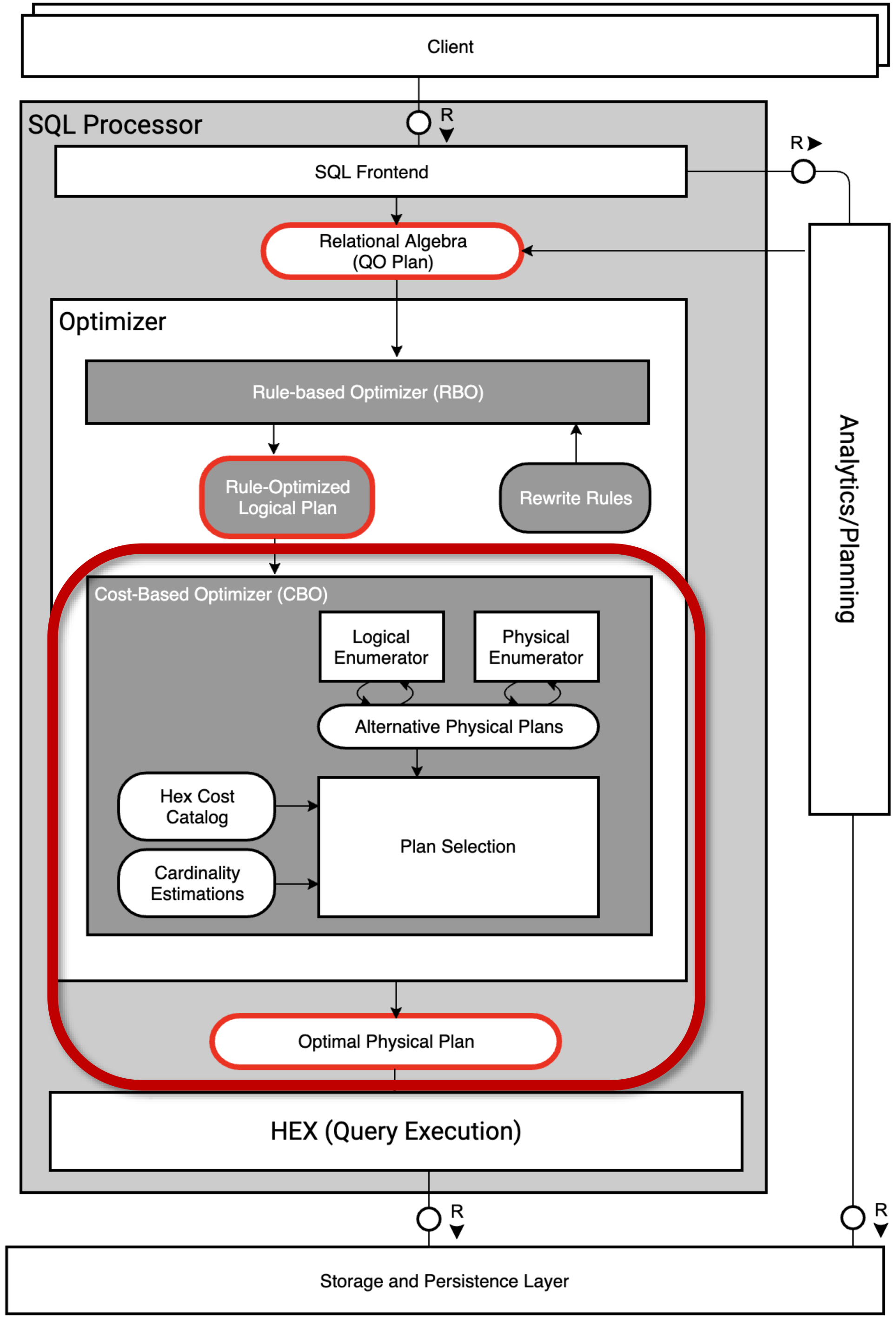




Optimizer (Cost-based)

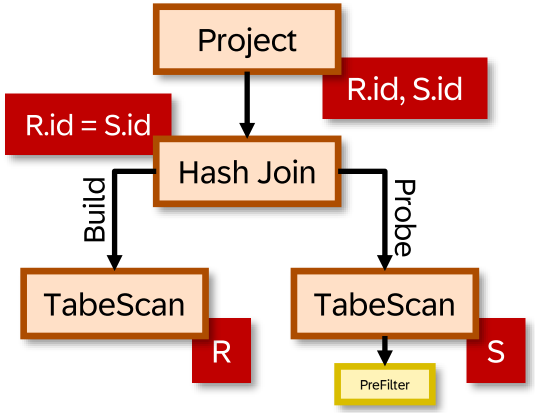
Rule-based optimized Plan





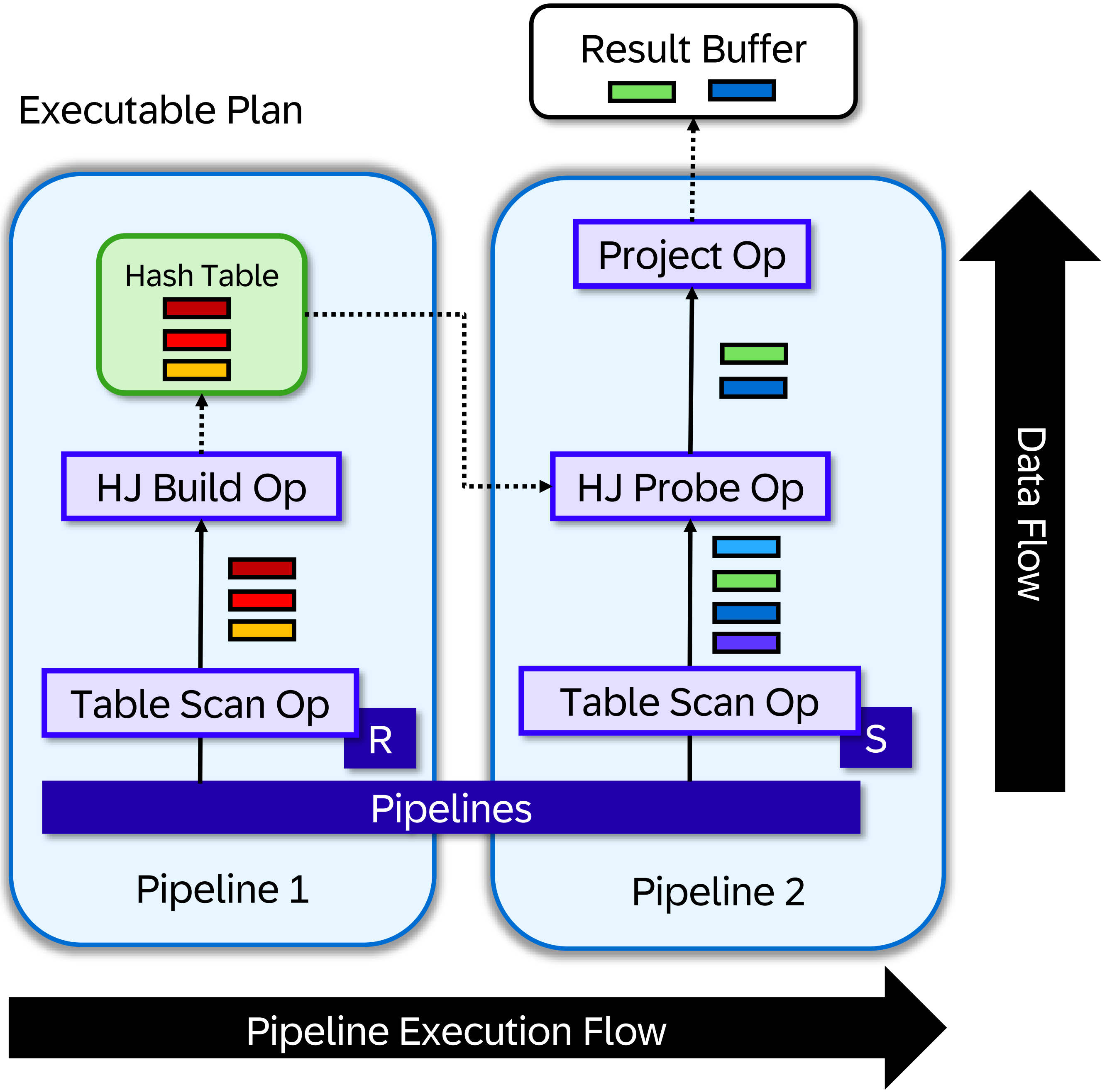
HEX Execution

Optimal Physical Plan

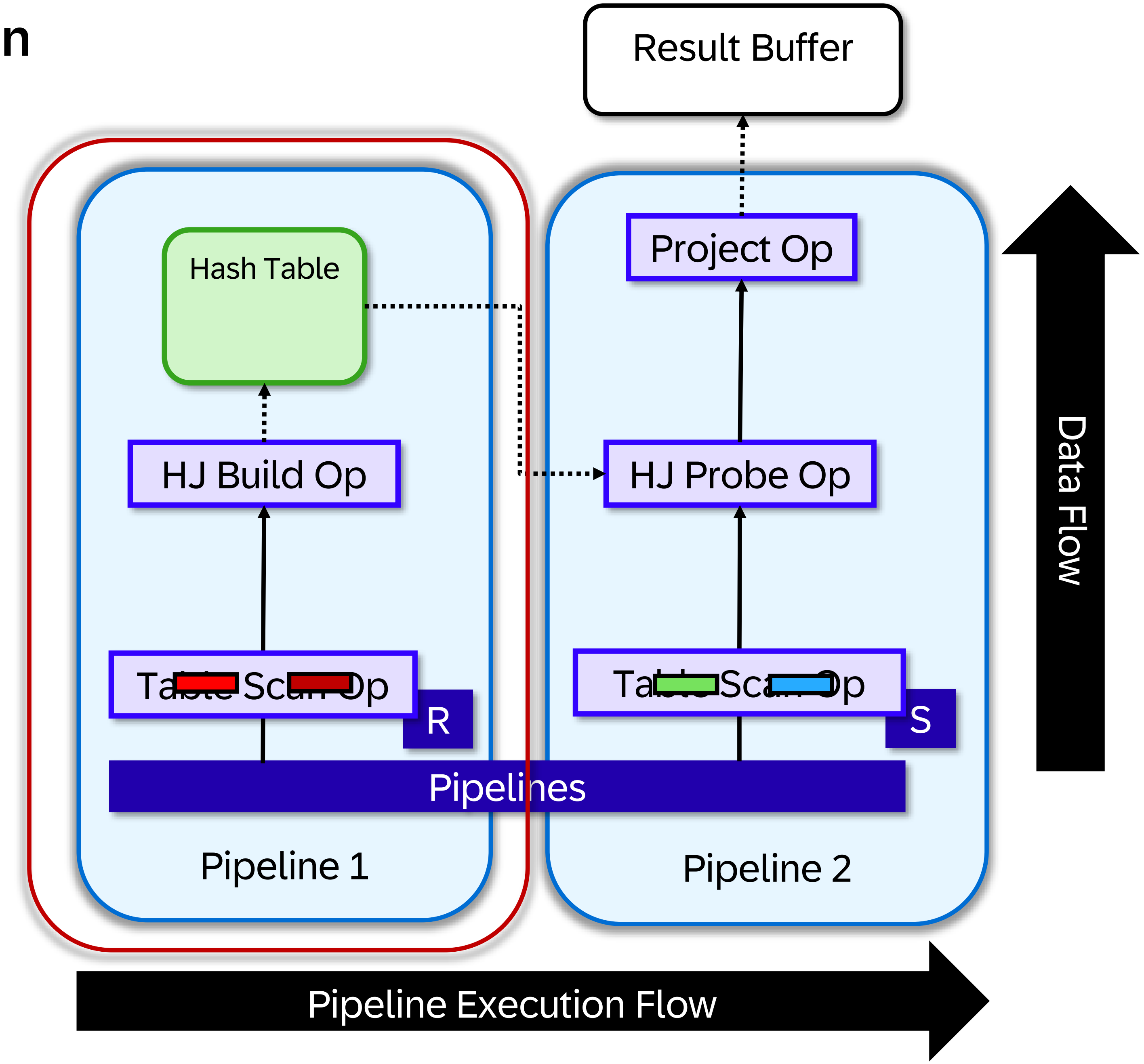


HEX Plan Generation

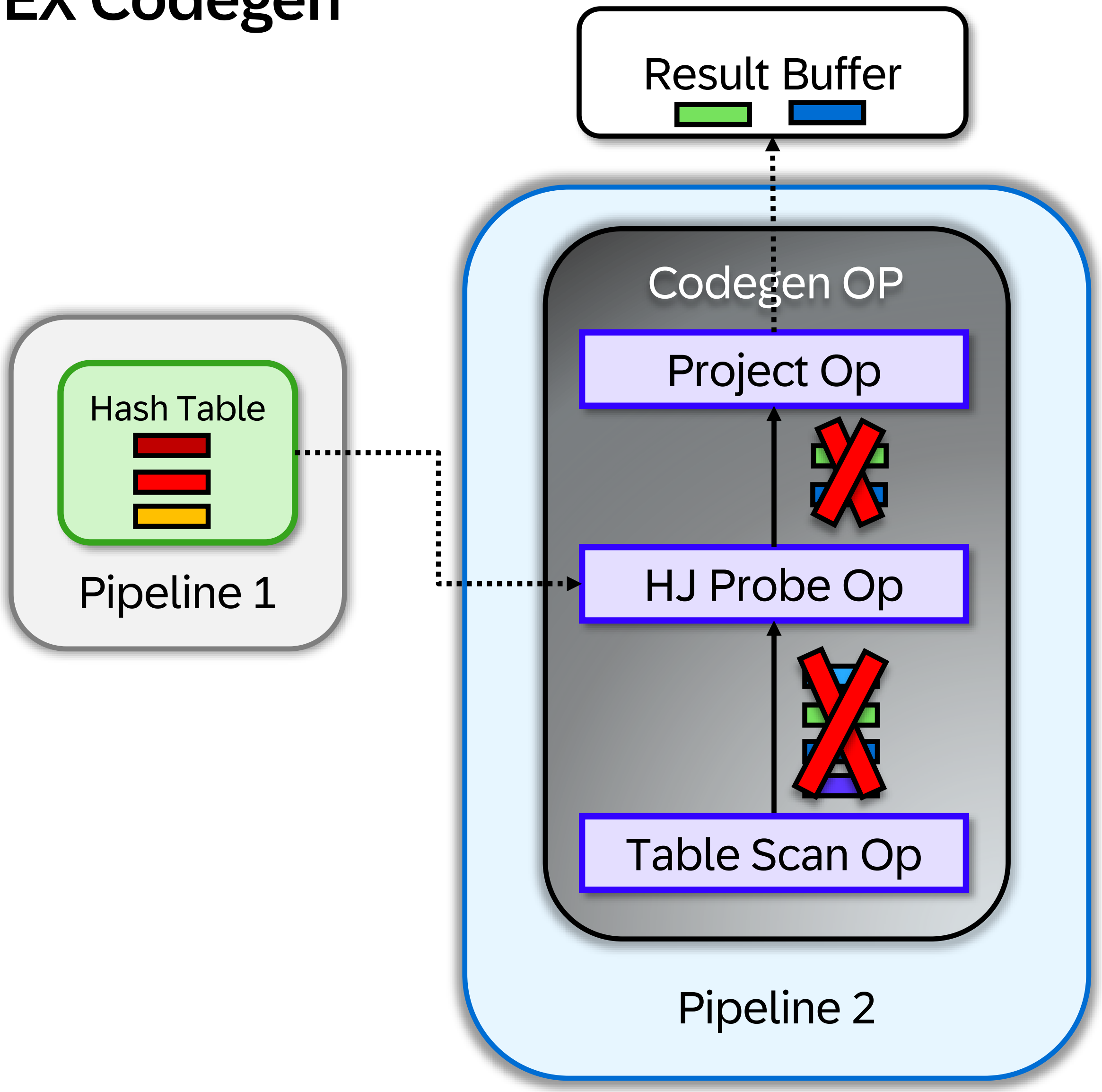
Executable Plan



HEX Execution

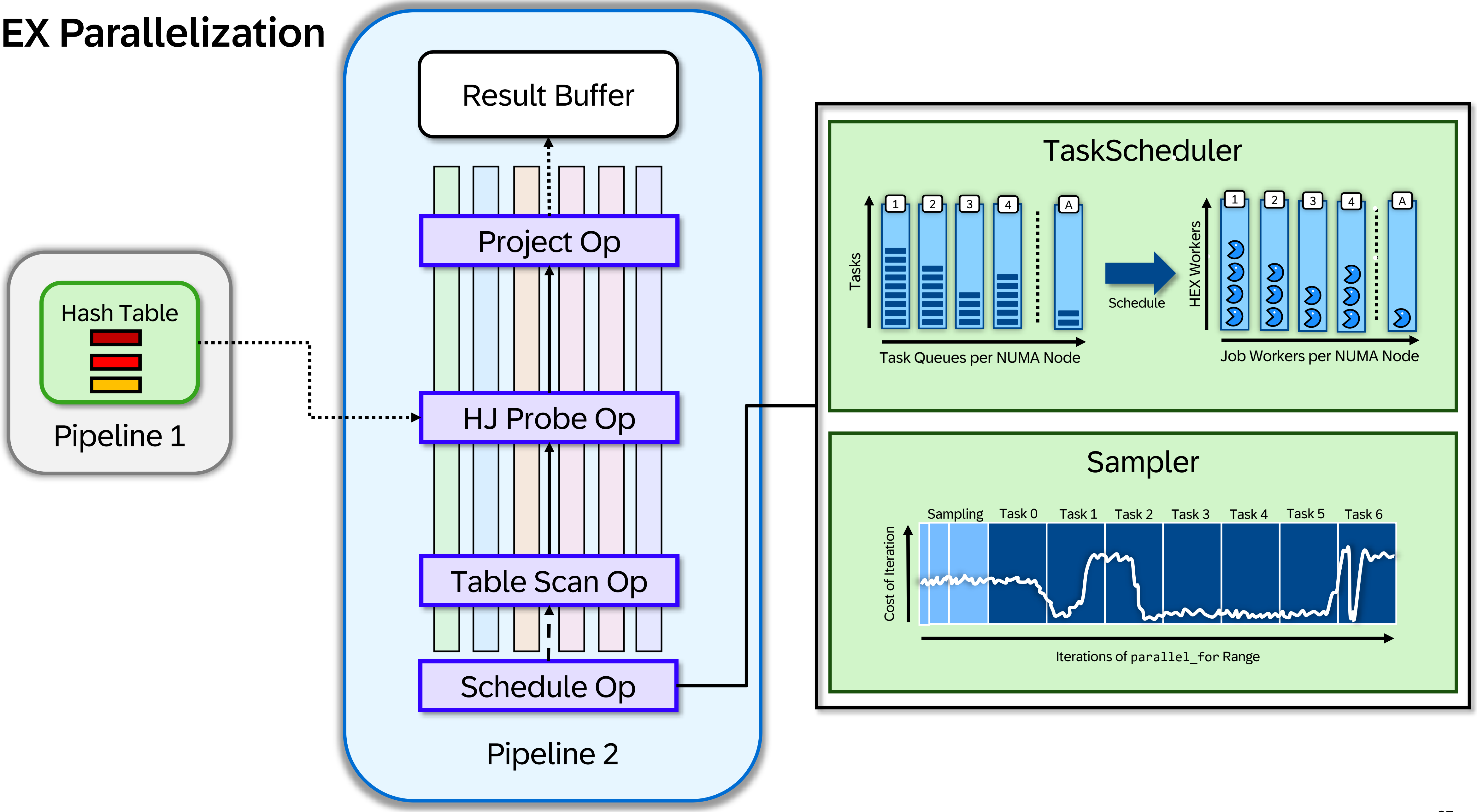


HEX Codegen



- **Reduced Interpretation Overhead**
- **Inlining and Specialization**
- **Improved Cache Locality**

HEX Parallelization



Recommendations

What to Look For in Case of Regressions

Performance

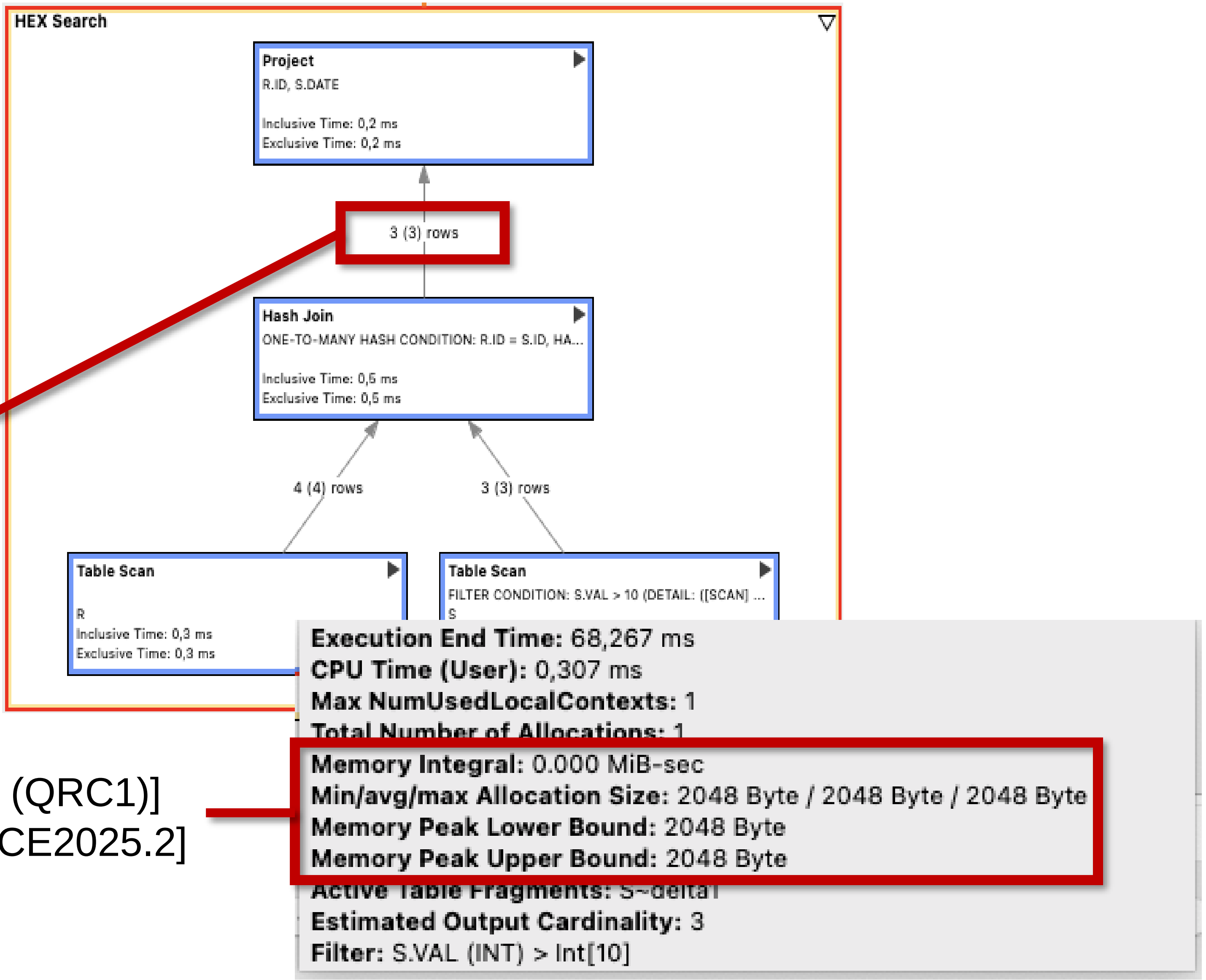
Cardinalities influence

- Algorithms
- Join Order
- Reparallelization
- JIT Compilations
- …

Hint: NO_HEX_FUSE_JIT_CODE

Memory

Hint: HEX_MEMORY_STATISTICS [>= CE2025.2 (QRC1)]
Config: [hex] telemetry_allocator_mode = 1 [< CE2025.2]



Troubleshooting

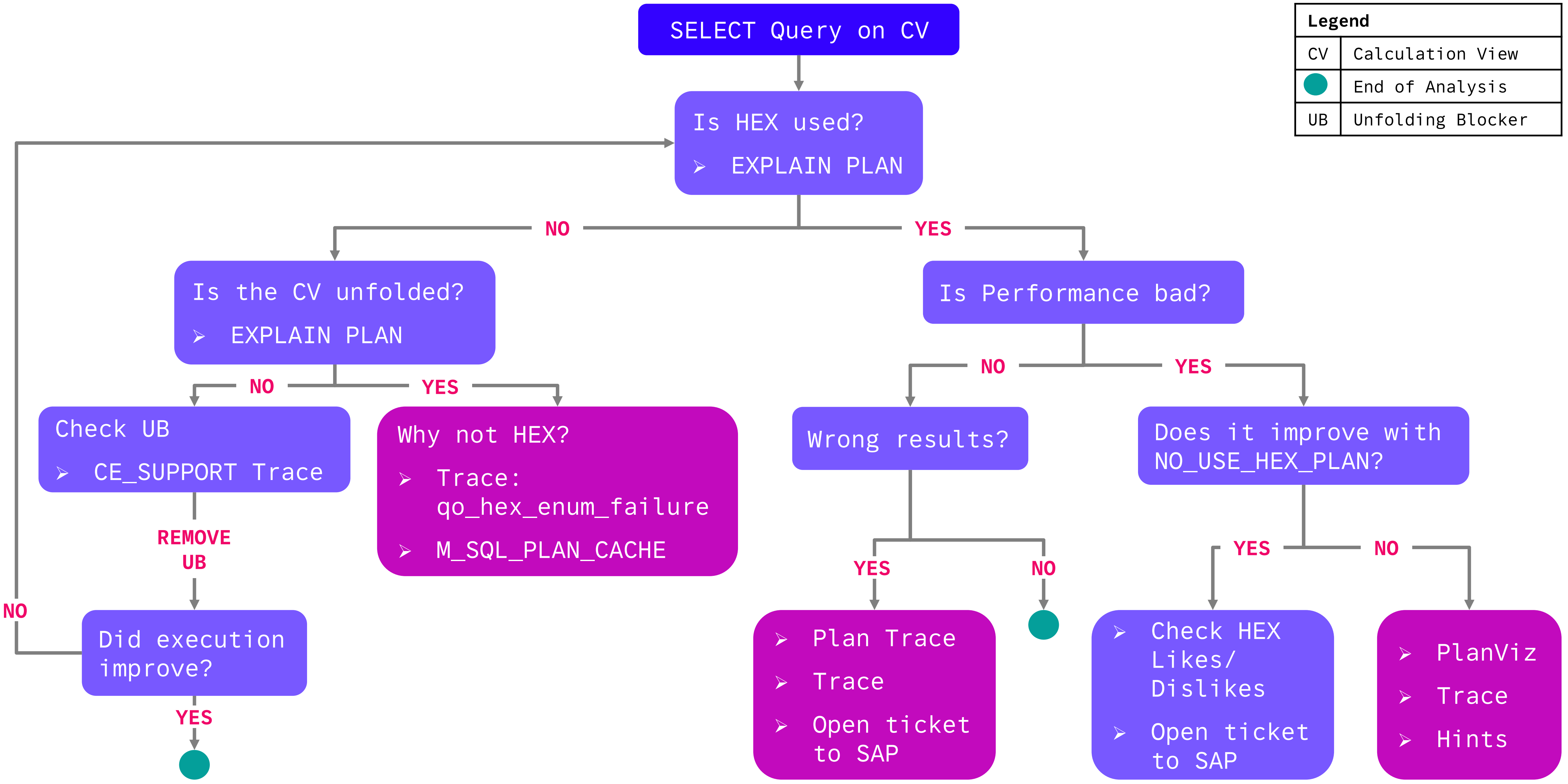
Troubleshooting

How to Analyze SQL Optimizer / HEX Decisions

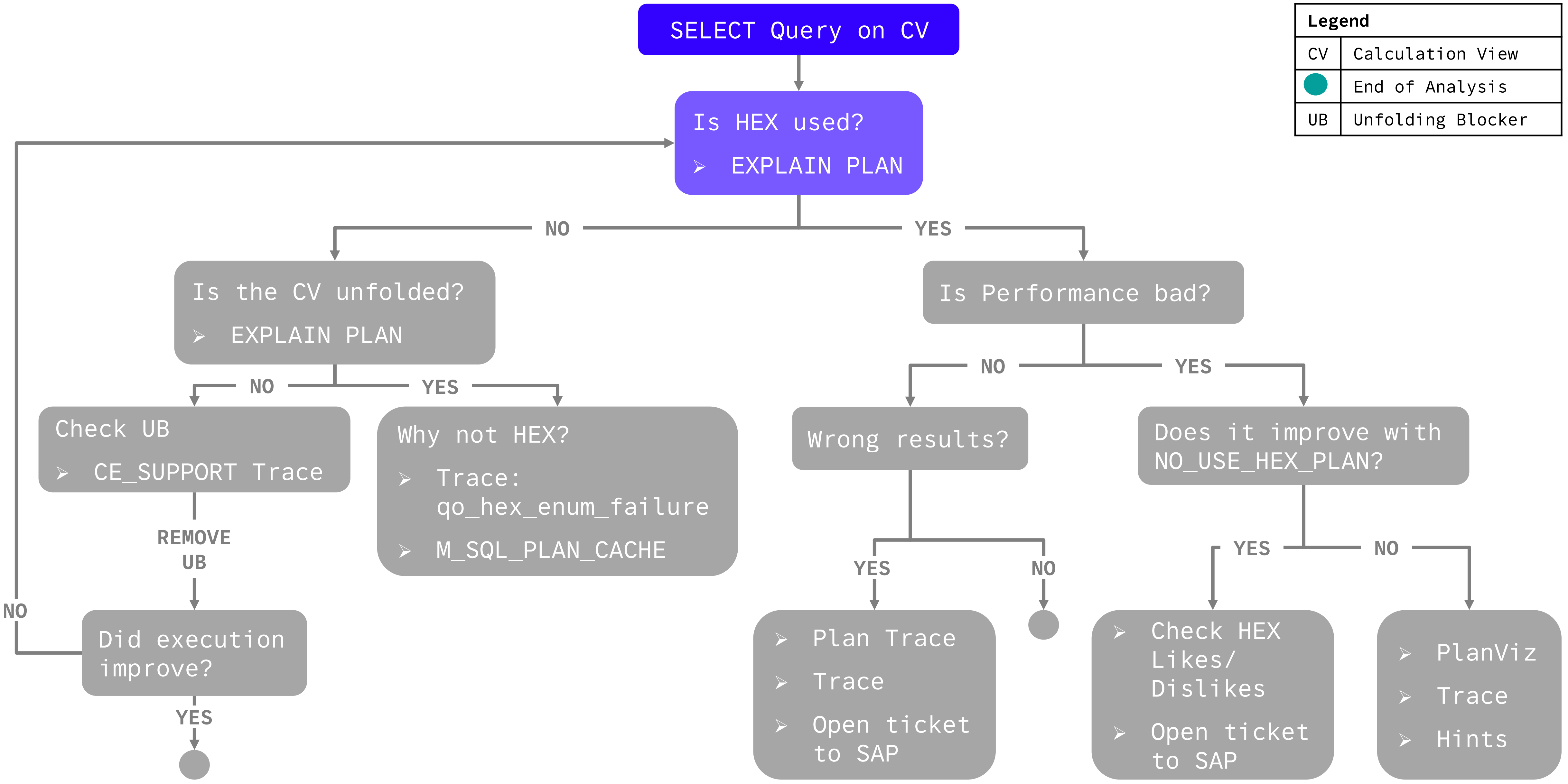
- EXPLAIN PLAN
- PlanViz / SQL Analyzer
- Tracing

How to Influence SQL Optimizer Decisions

- Hints
- (Remodeling)



Legend	
CV	Calculation View
●	End of Analysis
UB	Unfolding Blocker

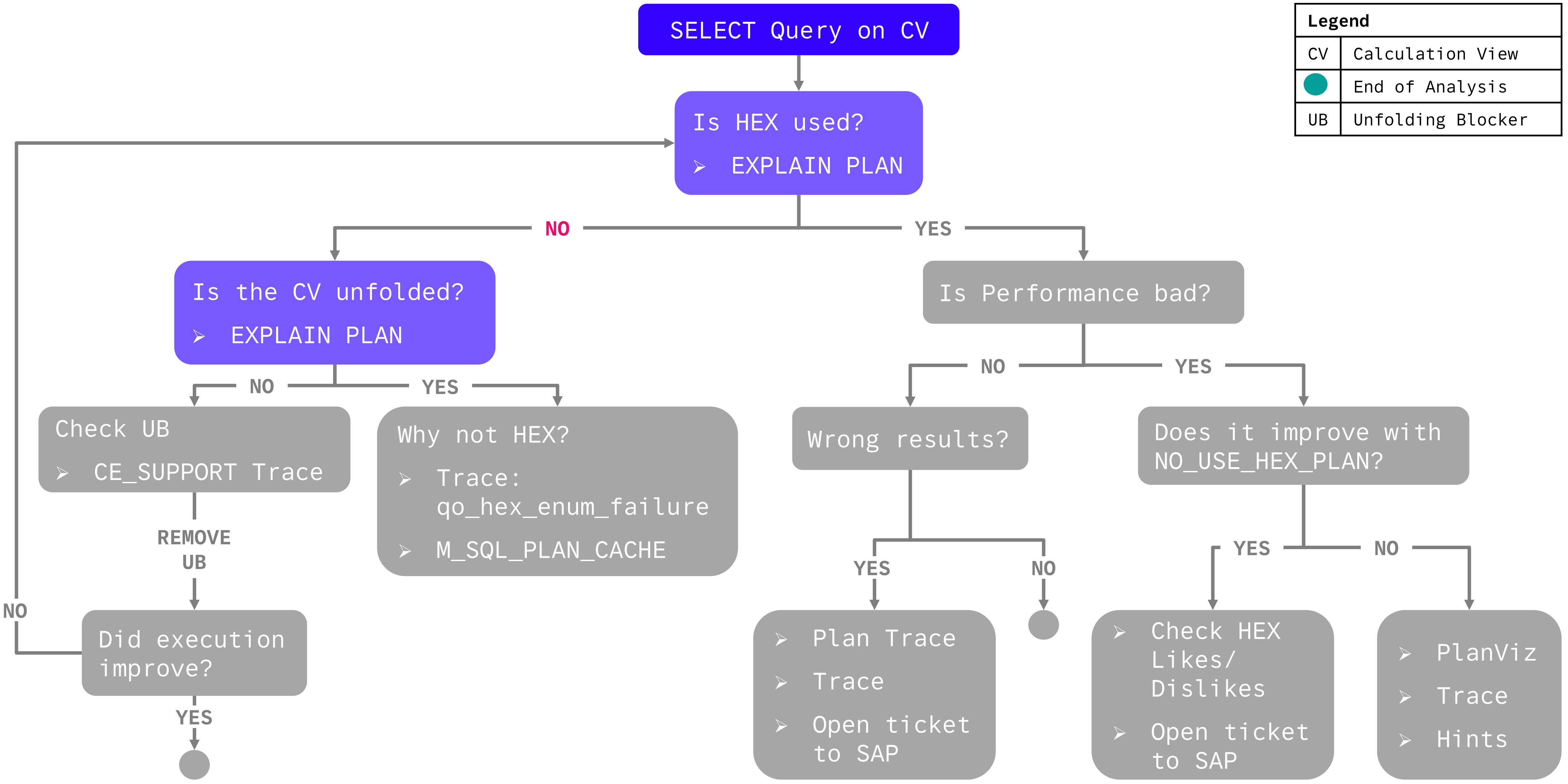


Legend

CV	Calculation View
●	End of Analysis
UB	Unfolding Blocker

Is HEX Used?

	OPERATOR_NAME	OPERATOR_DETAILS	OPERATOR_PROPERTIES	EXECUTION_ENGINE	DATABASE_NAME	SCHEMA_NAME
1	PROJECT	BIMC_VARIABLE_VIEW_INT.CATALOG_NAME, BIMC_VARIABLE_VIEW_INT.CUBE_NAME, BIM...	ATTACHED HINT LIST (USE_HEX_PLAN, OPTIMIZATION_LEVEL(RULE_BASED)), PHYSICAL_ENUM_BY: HEX_PROJECT	HEX	?	?
2	ORDER BY	BIMC_VARIABLE_VIEW_INT.ORDER ASC	PHYSICAL_ENUM_BY: HEX_ORDER_BY	HEX	?	?
3	HASH JOIN	HASH CONDITION: VIEWS.SCHEMA_NAME = C.SCHEMA_NAME AND VIEWS.VIEW_NAME ...	PHYSICAL_ENUM_BY: HEX_HASH_JOIN	HEX	?	?
4	UNION ALL	(CS_JOIN_INDEXES_.SCHEMA_NAME, CS_CALC_INDEXES_.SCHEMA_NAME, CS_HIERARCH...	PHYSICAL_ENUM_BY: HEX_UNION_ALL	HEX	?	?
5	TABLE SCAN		PHYSICAL_ENUM_BY: HEX_RS_TABLE_SCAN	HEX		SYS
6	TABLE SCAN	FILTER CONDITION: CS_CALC_INDEXES_.VIEW_NAME NOT LIKE '_SYS%' ESCAPE '\'	PHYSICAL_ENUM_BY: HEX_RS_TABLE_SCAN	HEX		SYS
7	TABLE SCAN		PHYSICAL_ENUM_BY: HEX_RS_TABLE_SCAN	HEX		SYS
8	HASH JOIN	HASH CONDITION: BIMC_VARIABLE_VIEW_INT.CATALOG_NAME = BIMC_AUTH_ONLY_CU...	TRANSLATION TABLE (LEFT): BIMC_VARIABLE_VIEW_INT.CATALOG_NAME = BIMC_AUTH_ONLY_CUBES.CATALOG_NAME, BIMC_...	HEX	?	?
9	INDEX JOIN (LEFT OUTER)	INDEX JOIN CONDITION: __cs_concat__(BIMC_VARIABLE_VIEW_INT.SCHEMA_NAME, BIMC...	PHYSICAL_ENUM_BY: HEX_INDEX_JOIN	HEX		_SYS_BI
10	INDEX JOIN (LEFT OUTER)	INDEX JOIN CONDITION: __cs_concat__(BIMC_VARIABLE_VIEW_INT.SCHEMA_NAME, BIMC...	PHYSICAL_ENUM_BY: HEX_INDEX_JOIN	HEX		_SYS_BI
11	INDEX JOIN	INDEX JOIN CONDITION: BIMC_VARIABLE_ASSIGNMENT.$trexexternalkey$ BETWEEN __cs_...	PHYSICAL_ENUM_BY: HEX_INDEX_JOIN	HEX		_SYS_BI
12	TABLE SCAN	FILTER CONDITION: BIMC_VARIABLE_VIEW_INT.IS_INPUT_ENABLED = 1 (DETAIL: ([SCAN] BI...	PHYSICAL_ENUM_BY: HEX_TABLE_SCAN	HEX		_SYS_BI
13	TABLE SCAN		PHYSICAL_ENUM_BY: HEX_TABLE_SCAN	HEX		_SYS_BI


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

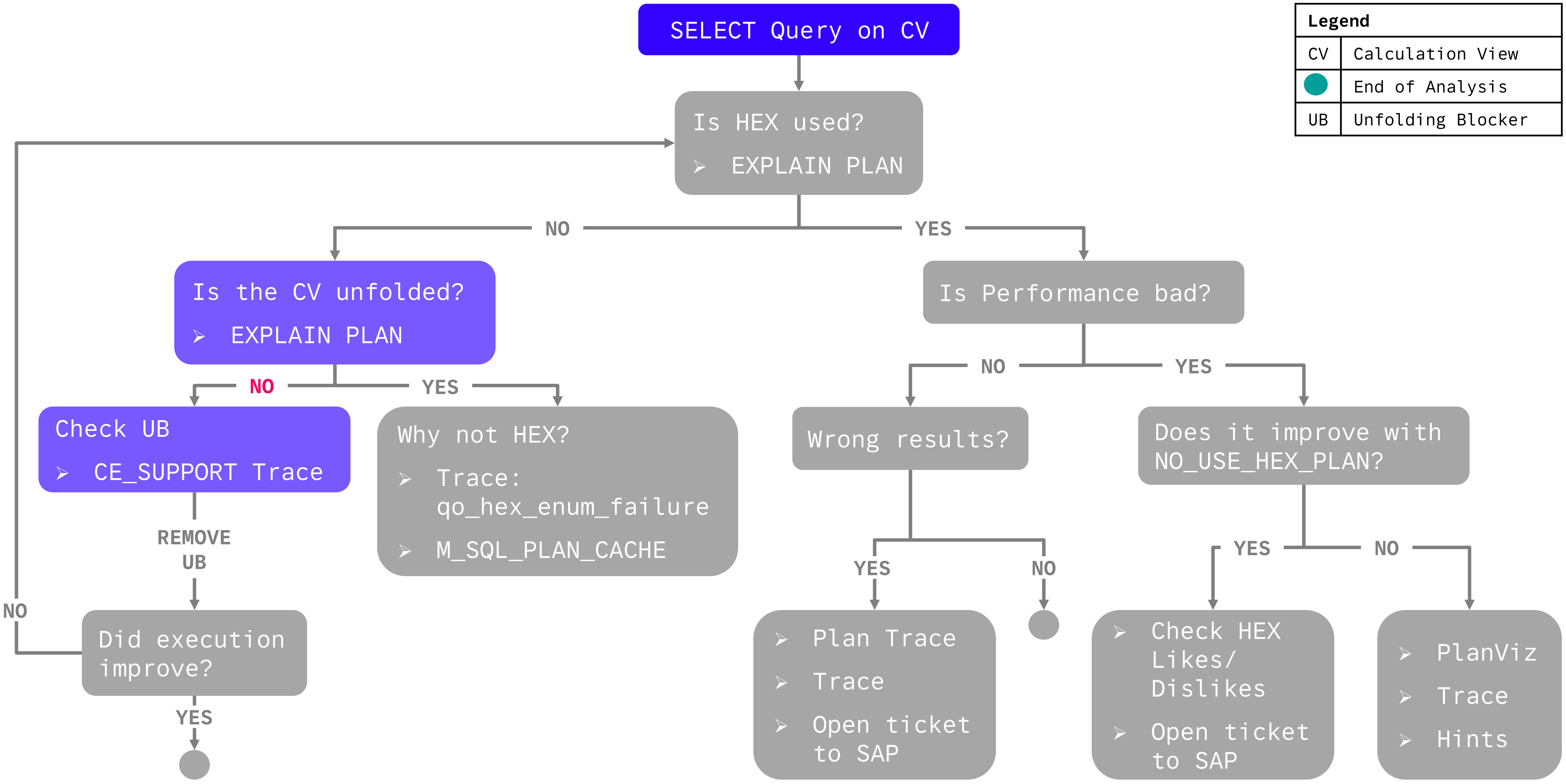
Is the Calculation View Unfolded?

Rows (2)

	OPERATOR_NAME	OPERATOR_DETAILS	OPERATOR_PROPERTIES	EXECUTION_ENGINE	DATABASE_NAME	SCHEMA_NAME	TABLE_NAME	TABLE_TYPE
1	PROJECT	CV_BOOKS.TITLE, CV_BOOKS.A...	PHYSICAL_ENUM_BY: HEX_PROJECT	HEX	*NULL*	*NULL*	*NULL*	*NULL*
2	CALCULATION VIEW			CALC	H00	HANA_TEC_CON_2025_HDI_DB01_1	CV_BOOKS	CALCULATION VIEW

Rows (2)

	OPERATOR_NAME	OPERATOR_DETAILS	OPERATOR_PROPERTIES	EXECUTION_ENGINE	DATABASE_NAME	SCHEMA_NAME	TABLE_NAME	TABLE_TYPE
1	COLUMN SEARCH	CV_BOOKS.TITLE, CV_BOOKS.A...	LATE MATERIALIZATION, ATTACHED...	COLUMN	*NULL*	*NULL*	*NULL*	*NULL*
2	COLUMN VIEW			COLUMN	H00	HANA_TEC_CON_2025_HDI_DB01_1	CV_BOOKS	CALCULATION VIEW

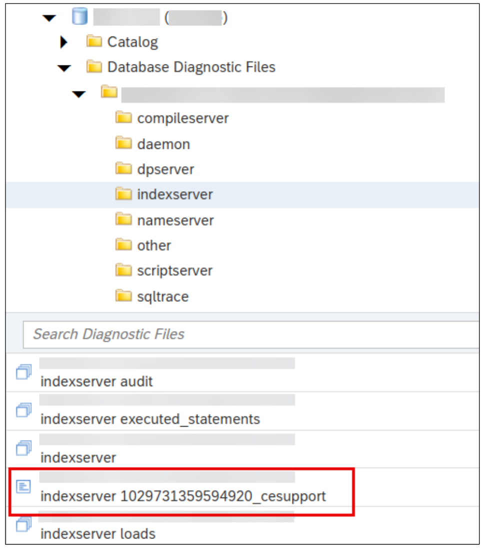


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

Calculation View not Unfolded: Check Unfolding Blocker

CE_SUPPORT Trace

```
--<QUERY>
WITH PARAMETERS('PLACEHOLDER' = ('$$CE_SUPPORT$$', 'd_calcengine, d_ceinstantiate, d_ceoptimizer, i_ceoptimizerinfo, d_ceqo, d_ceplanexec, d_ceexecute'))
```



Unsupported function:

abap_lower()

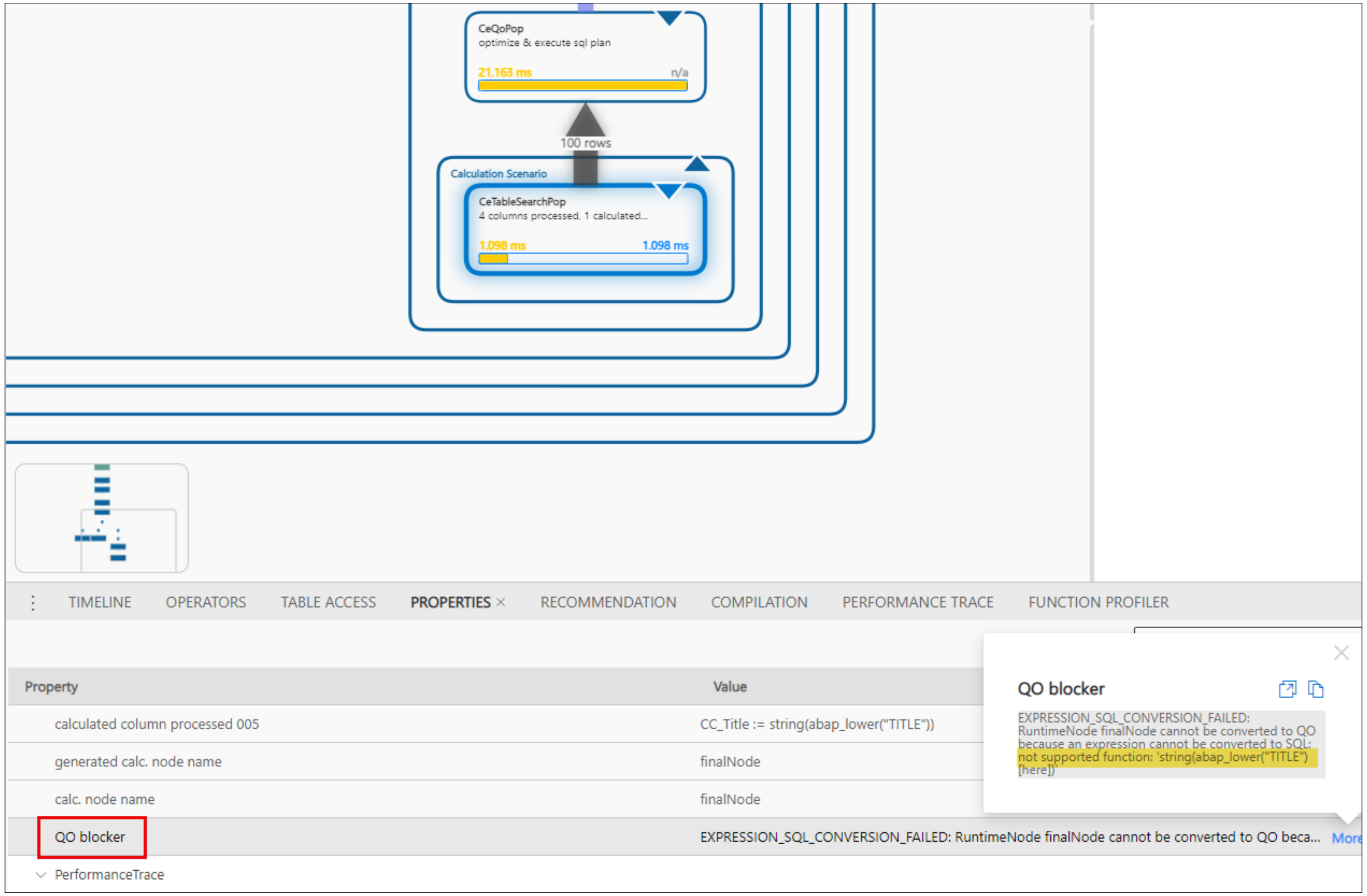
Look for **NodeIsQOCompliant** in trace!

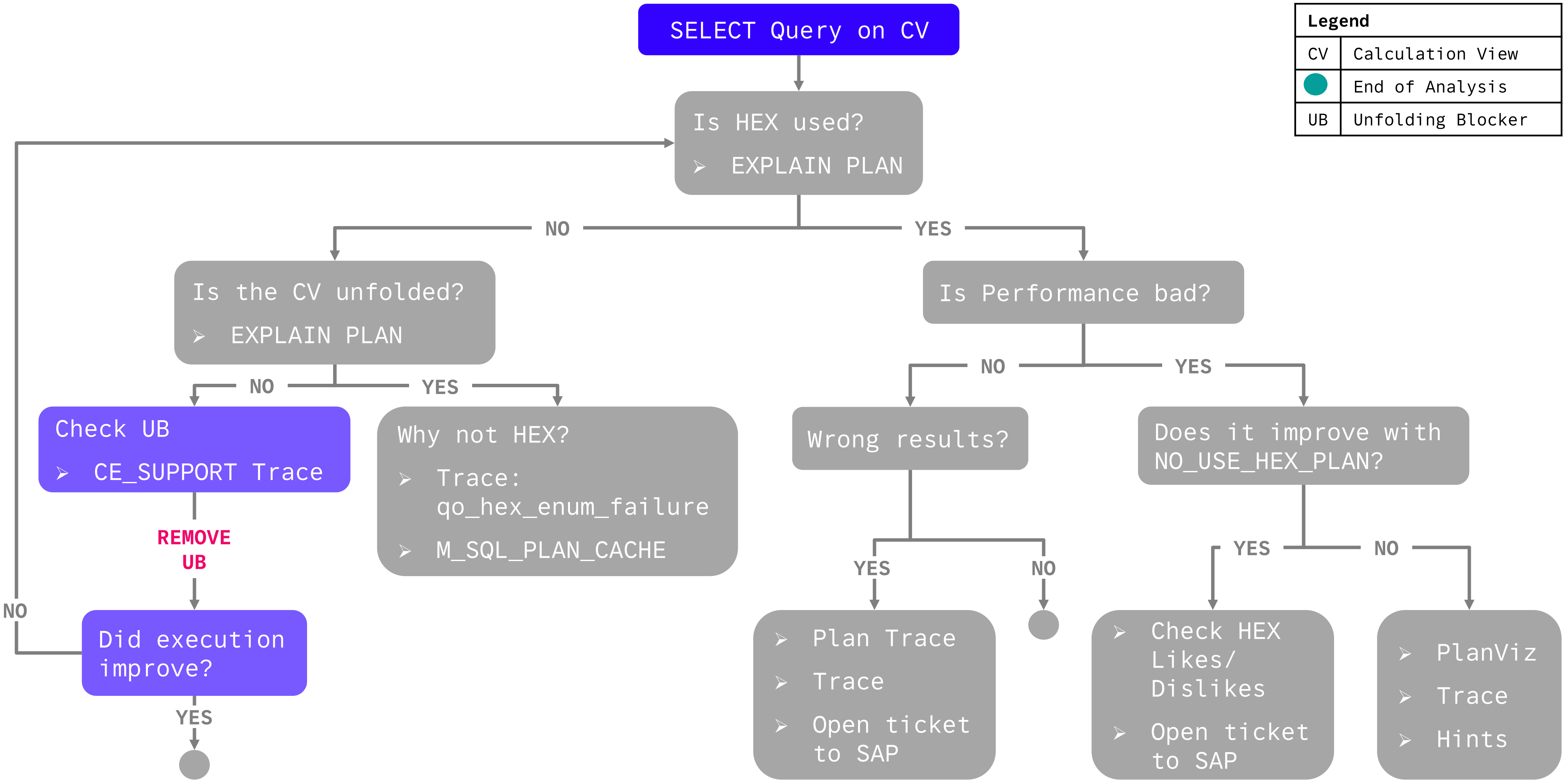
```
[12751]{208236}[249/-1]<197ee6a8f89dd3382a8fd1d52b4219dc#fb4c2401ea980a89> 2025-07-09 09:01:57.634309 i ceQo          ceNodeIsQOCompliantPattern.cpp(00133) : QO
    unfolding/QO conversion is not possible. Tag: EXPRESSION_SQL_CONVERSION_FAILED Reason: RuntimeNode finalNode cannot be converted to QO because an expression cannot
    be converted to SQL: not supported function: 'string(abap_lower("TITLE")[here])'
```

```
[62672]{239753}[1237/-1]<197d173884c8d3a72251f1ff1494f221#41e92e5ca70ed348> 2025-07-03 18:15:30.292085 i ceOptimizer      ceOptStruct.cpp(00039) : Pattern
    NODE_IS_QO_COMPLIANT_PATTERN(36) does match for node '$REQUEST$' op(AGGREGATION_OP)
[62672]{239753}[1237/-1]<197d173884c8d3a72251f1ff1494f221#41e92e5ca70ed348> 2025-07-03 18:15:30.292088 i ceOptimizer      ceOptimizer2.cpp(00819) : Rule
    NODE_IS_QO_COMPLIANT_RULE(34) was applied for node '$REQUEST$' op(AGGREGATION_OP).
[62672]{239753}[1237/-1]<197d173884c8d3a72251f1ff1494f221#41e92e5ca70ed348> 2025-07-03 18:15:30.292251 i ceOptimizer      ceNodeIsQOCompliantPattern.cpp(00129) :
    NodeIsQOCompliantPattern::isConversionOfCalcAttrsToQoPossible: node 'finalNode' op(PROJECTION_OP) has error: exception  1: no.1000013  (ptime/query/plan_generator
    /dml_search/expression/ConvertExpression.cpp:774) TID: 62672
    not supported function: 'string(abap_lower("TITLE")[here])'
exception throw location:
```

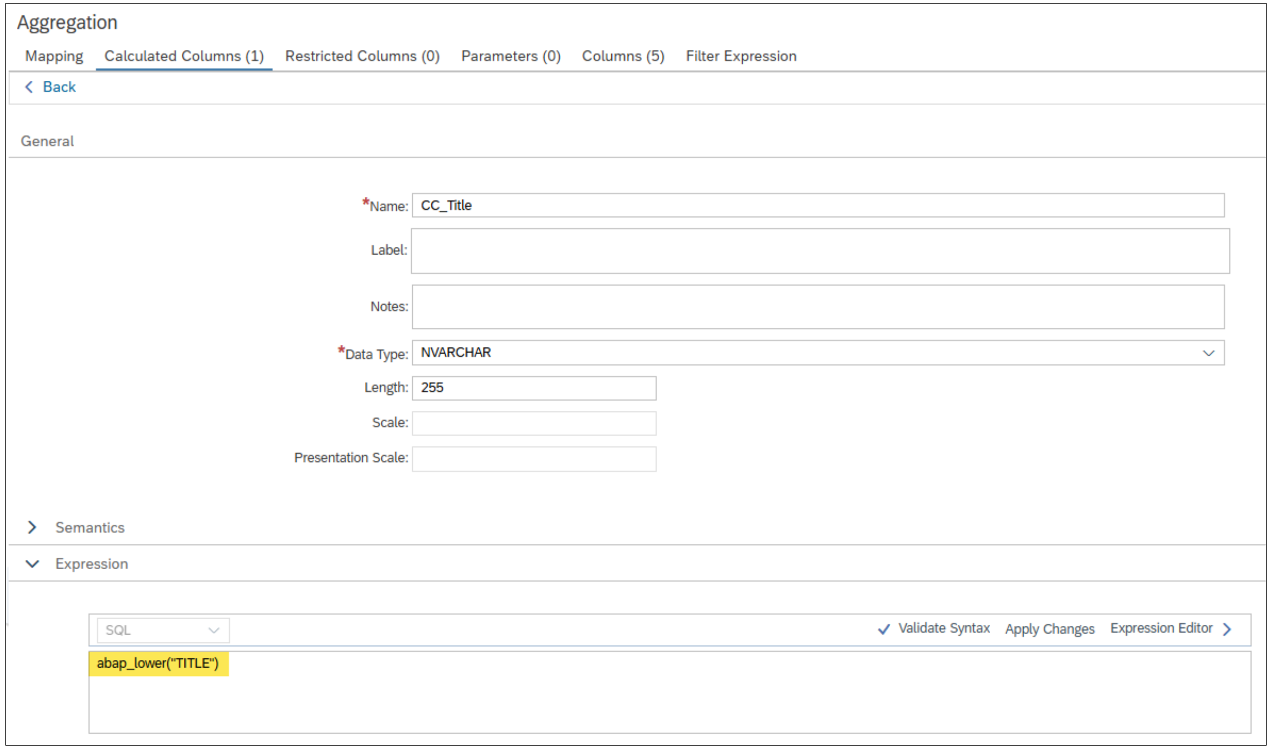
Calculation View not Unfolded: Check Unfolding Blocker

PlanViz




Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker
SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints

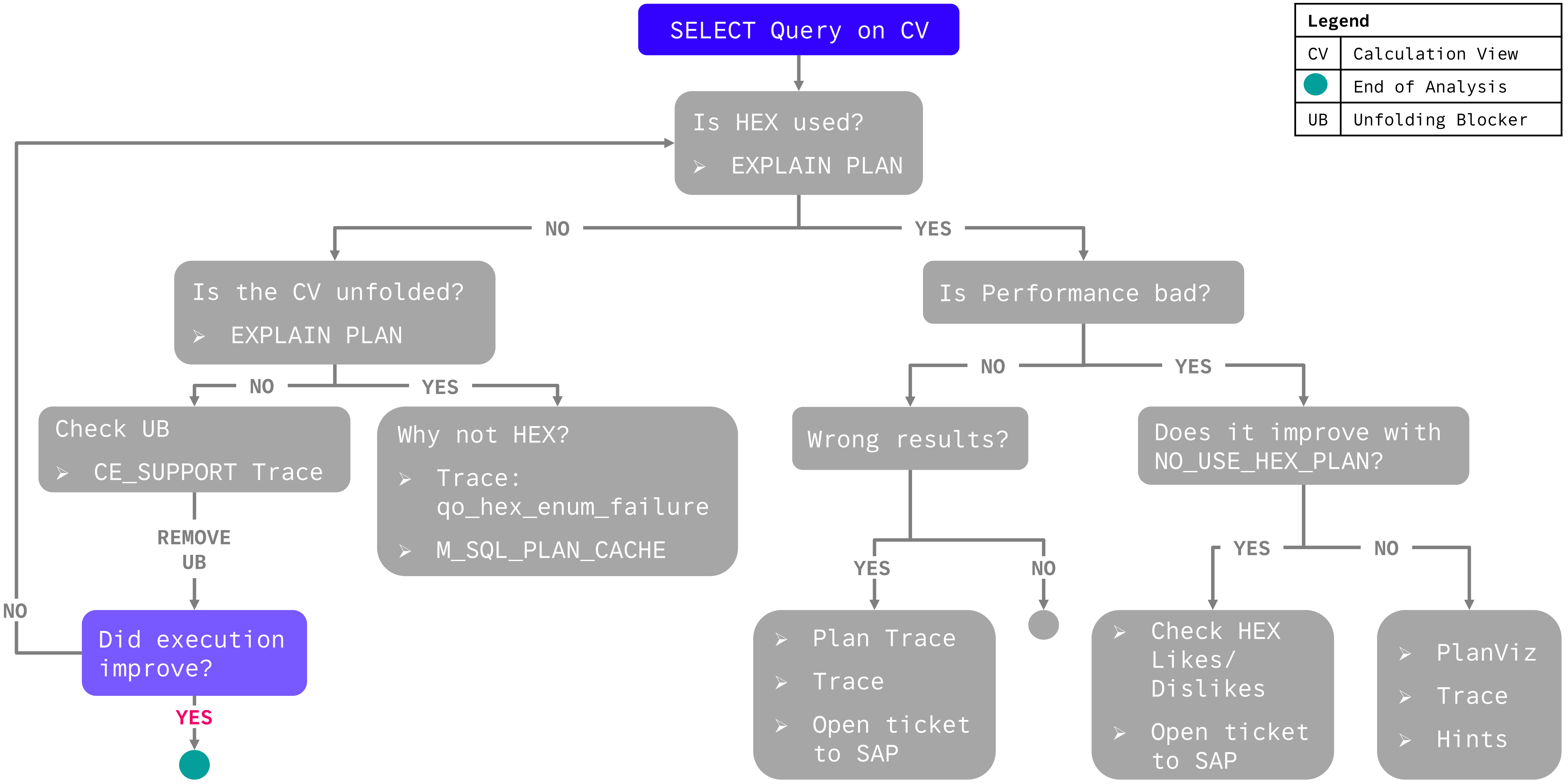
Remove the Unfolding Blocker

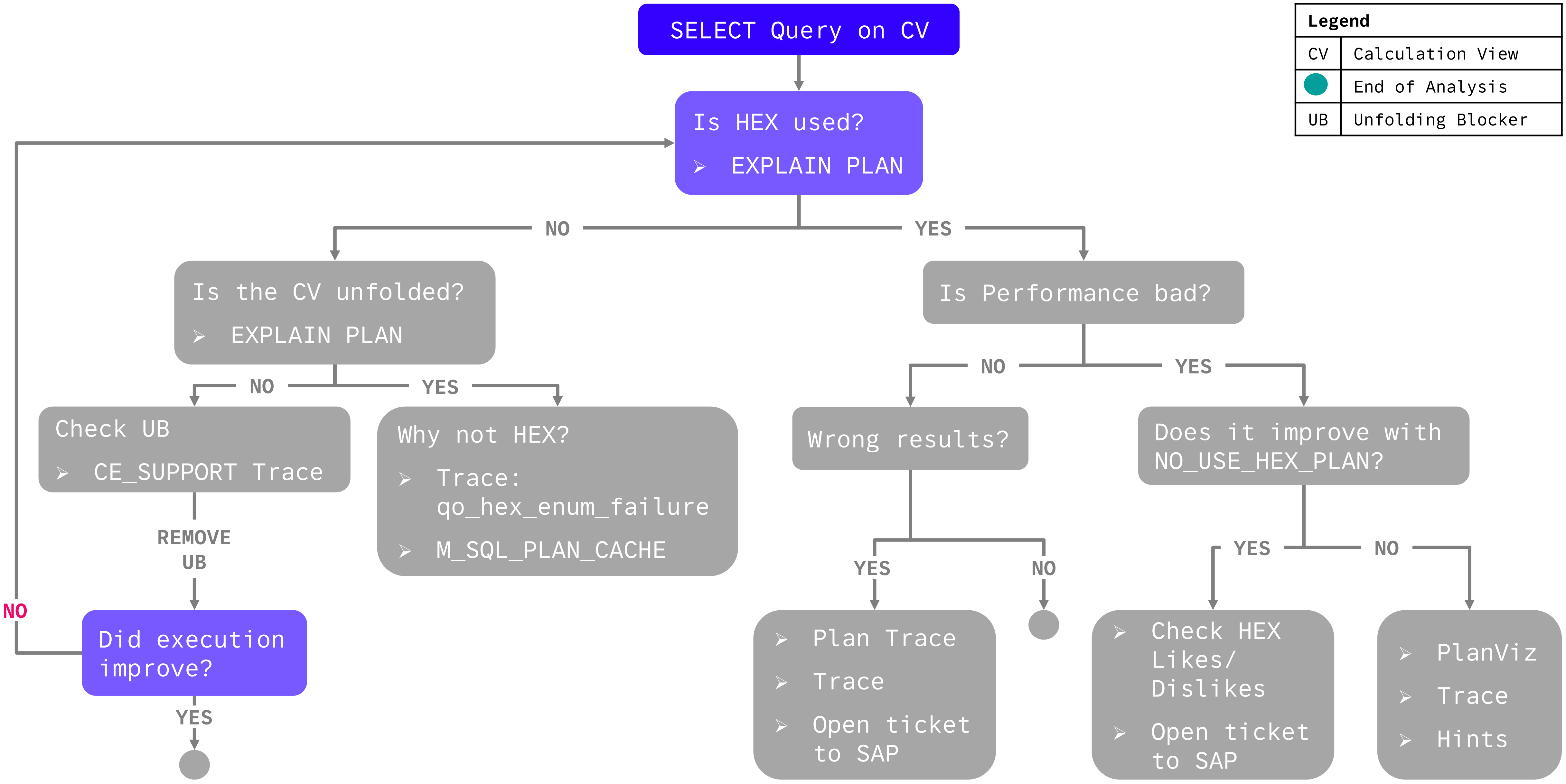


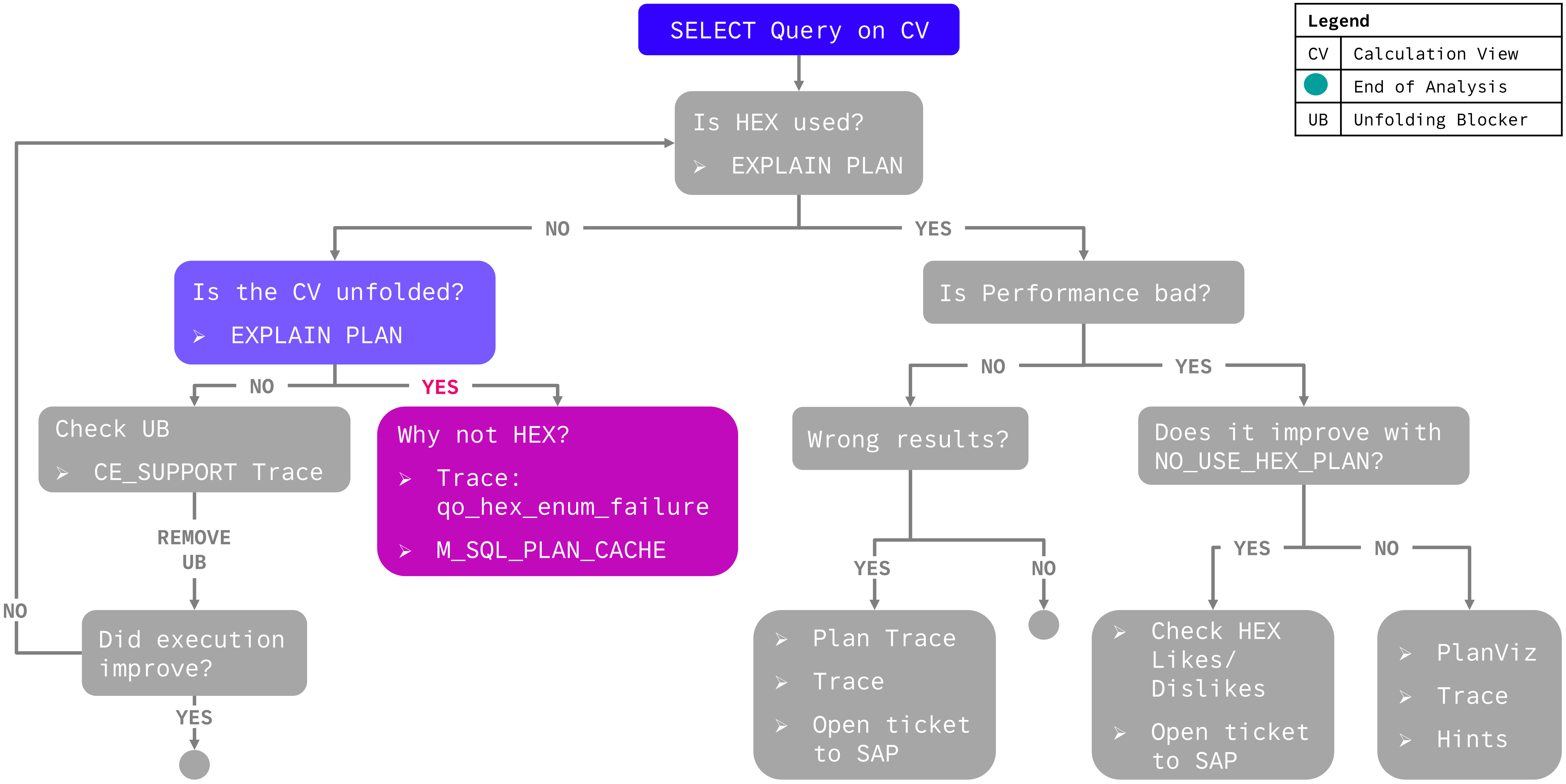
Look for **successfully unfolded** in trace!

```
[3529]{208236}[249/-1]<197ee6a8f89dd3382a8fd1d52b4219dc#fb4c2401ea980a89> 2025-07-09 09:28:36.286052 d CalcEngine       ceController2.cpp(01566) : View
    'HANA_TEC_CON_2025_HDI_DB01_1:CV_BOOKS_UNFOLDED' successfully unfolded
```

```
[40160]{239753}[2542/-1]<197d173884c8d3a72251f1ff1494f221#41e92e5ca70ed348> 2025-07-03 18:37:54.205904 i ceOptimizer      ceOptStruct.cpp(00039) : Pattern
    NODE_IS_QO_COMPLIANT_PATTERN(36) does match for node 'finalNode' op(PROJECTION_OP)
[40160]{239753}[2542/-1]<197d173884c8d3a72251f1ff1494f221#41e92e5ca70ed348> 2025-07-03 18:37:54.205906 i ceOptimizer      ceOptimizer2.cpp(00819) : Rule
    NODE_IS_QO_COMPLIANT_RULE(34) was applied for node 'finalNode' op(PROJECTION_OP).
```


Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker
SELECT Query on CV
Is HEX used?
EXPLAIN PLAN
NO
YES
Is the CV unfolded?
EXPLAIN PLAN
NO
YES
Check UB
CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
Trace: qo_hex_enum_failure
M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
Plan Trace
Trace
Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
Check HEX Likes/ Dislikes
Open ticket to SAP
PlanViz
Trace
Hints


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker


SELECT Query on CV
Is HEX used?
EXPLAIN PLAN
NO
YES
Is the CV unfolded?
EXPLAIN PLAN
NO
YES
Check UB
CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
Trace: qo_hex_enum_failure
M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
Plan Trace
Trace
Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
Check HEX Likes/ Dislikes
Open ticket to SAP
PlanViz
Trace
Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

Calculation View Unfolded: Why not HEX?

qo_hex_enum_failure

Analyze

```sql
ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM' ) SET
('traceprofile_why_not_hex', 'sql_user') = '<user>',
('traceprofile_why_not_hex', 'qo_hex_enum_failure') = 'debug'
WITH RECONFIGURE;

 -- QUERY

ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM' ) UNSET
('traceprofile_why_not_hex')
WITH RECONFIGURE;
```

Look for
qo_hex_enum_fail
in trace!

```
[31692]{239753}[3110/1837533757442]<197d173884c8d3a72251f1ff1494f221#41e92e5ca70ed348> 2025-07-03 19:08:15.830332 d qo_hex_enum_fail CapabilityChecker.cc(00158) : [rel: Root, opId:L:2]
    HEX Enumeration Failure: HEX is disabled because it is unavailable for dml (reason: tableChangeOperator is not set.)
```

```
[26347]{242474}[271979/-1]<18df535b6daf323989153378862399c2#0000000000000000> 2024-02-29 14:18:53.004126 d qo_hex_enum_fail GroupByRule.cc(00186) :
[rel: Group By, opId:L:1115] Hex Enumeration Failure: estimated input size for group by with certain formulas exceeds limit (hex_limit_input_cardinality_group_by_with_granularity) (1.14462e+07 vs. 100)
```

Calculation View Unfolded: Why not HEX?

M_SQL_PLAN_CACHE

Analyze | Current schema: | Connected to:

```sql
SELECT STATEMENT_STRING, STATEMENT_HASH, EXECUTION_ENGINE, HEX_REJECTION_REASON
FROM M_SQL_PLAN_CACHE
WHERE EXECUTION_ENGINE != 'HEX'
AND HEX_REJECTION_REASON != '';
```

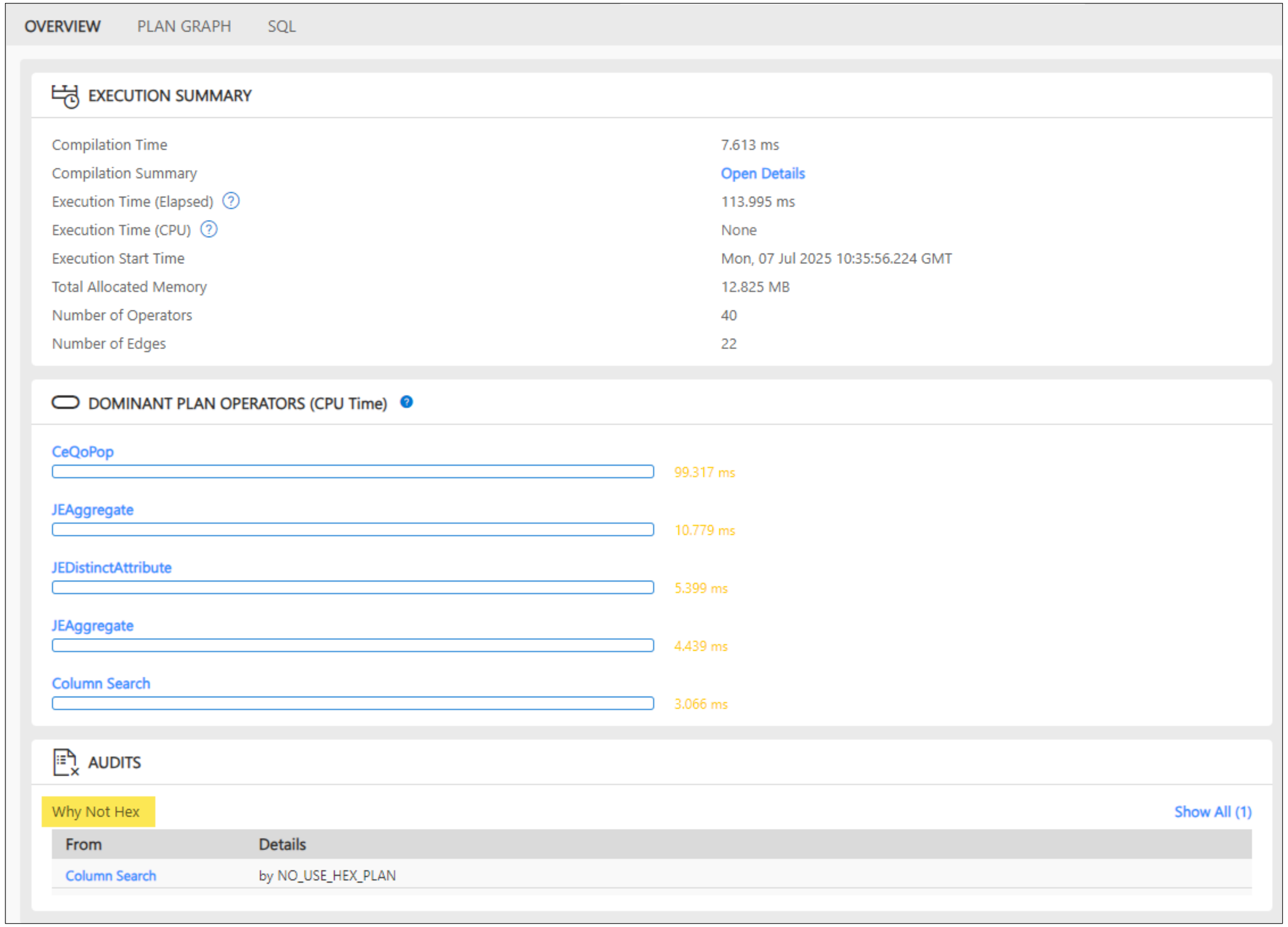
Result x | Messages x | History

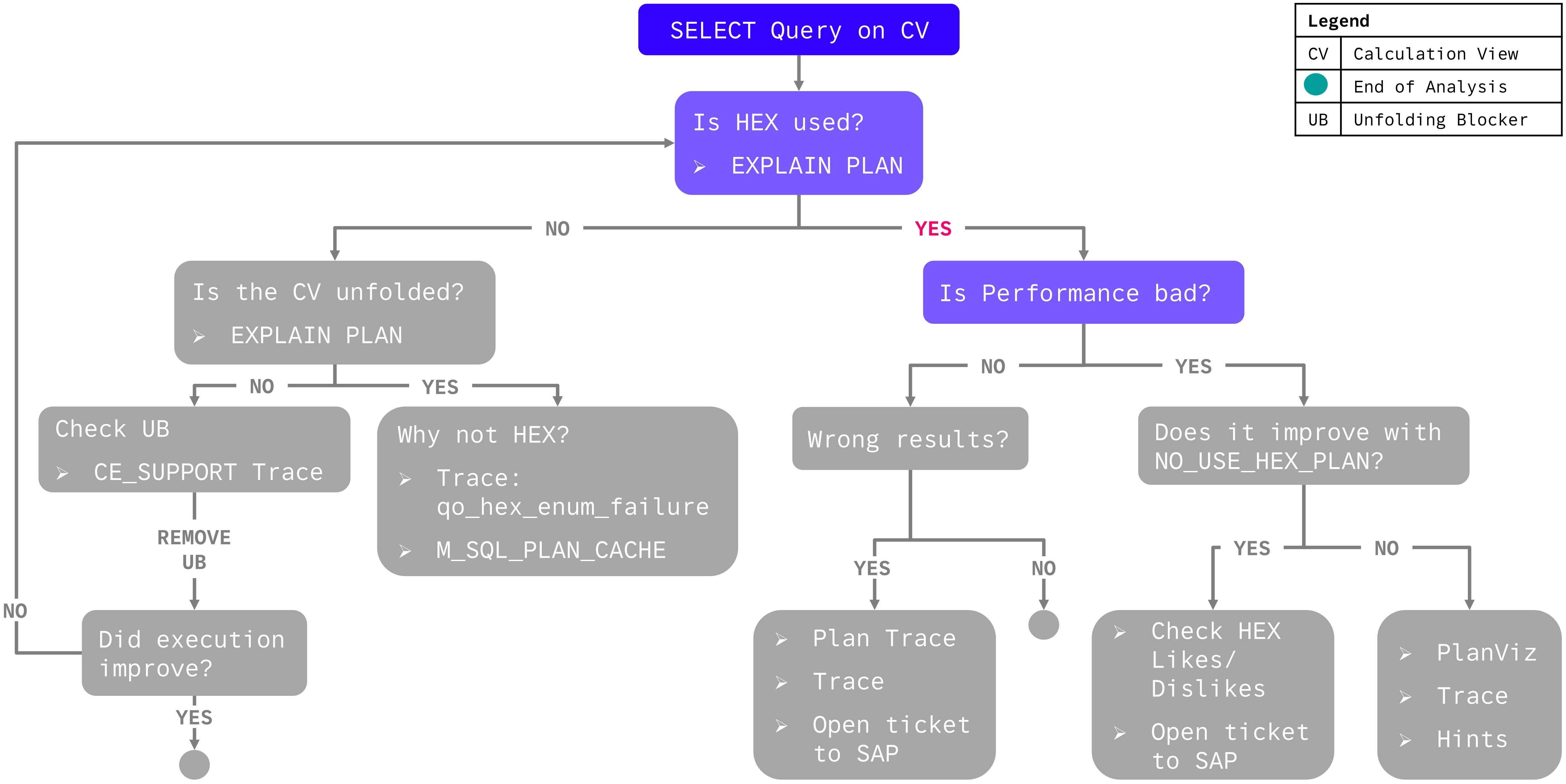
Rows (1011) | SQL

	STATEMENT_STRING	STATEMENT_HASH	EXECUTION_ENGINE	HEX_REJECTION_REASON
1	/* procedure: "SAP_PA_APL"."sap.pa.apl.base.internal::L...	4c36a2e00c2e17bcb78106ad69c29f6d	COLUMN, ROW	not generally available for row table only plan
2	SELECT COUNT(*) FROM PSE_CERTIFICATES WHERE P...	6547904a2d05e966fc4936a9b84167cb	ROW	not generally available for row table only plan
3	SELECT EXCEPTION_OID FROM SYS.REMOTE_SUBSCR...	8cb66248015f0f310e9ad87726950fed	ROW	FALLBACK_USED option is given
4	select top 1 exception_oid from sys.remote_subscription_...	9113c3e977e6aa275a7fb13c06955540	ROW	FALLBACK_USED option is given
5	/* procedure: "_SYS_STATISTICS"."ALERT_PASSWORD_E...	b1f340a1fcd669c5afe18404942dbecb	ROW, ESX	not generally available for row table only plan
6	SELECT COUNT(DISTINCT u.USER_NAME) FROM SYS.U...	2061fd43d7e39e927c6f42b31e227695	ROW	not generally available for row table only plan
7	select * from (SELECT SCHEMA_NAME AS ...	80465c127505d0d3b1fd1ff1e09e68c6	ROW	not generally available for row table only plan
8	/* procedure: "_SYS_STATISTICS"."ALERT_CHECK_DEPR...	f595785783e8873af3827a3ec6eb7165	ROW, ESX	not yet available for dml (tableChangeOperator is not set)
9	SELECT CASE WHEN TYPE_NAME = 'TIMESTAMP' AND...	092fadc9d8bdb41dc30bc6fd71328482	ROW	not generally available for row table only plan
10	SELECT COUNT(*) FROM SYS.ESH_MODEL_PROPERTY...	acc7fd5fe495642f278d02e50faa0118	ROW	not generally available for row table only plan

Calculation View Unfolded: Why not HEX?

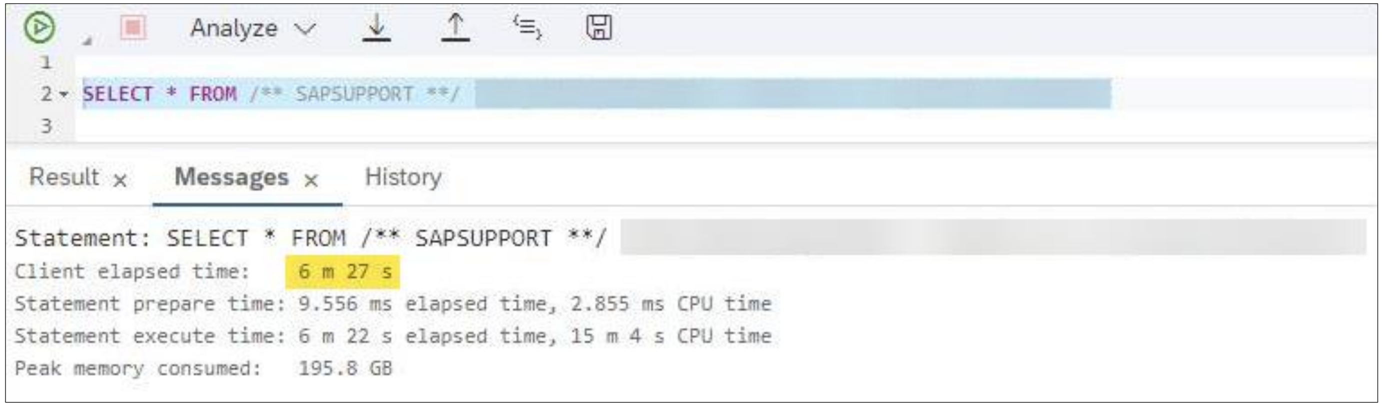
PlanViz / SQL Analyzer

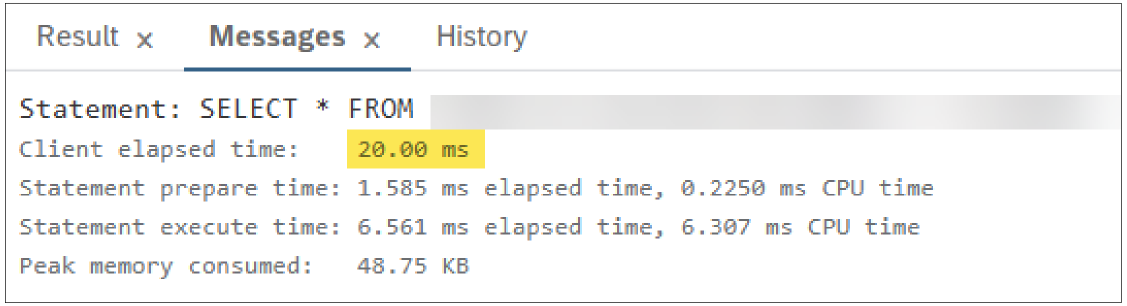


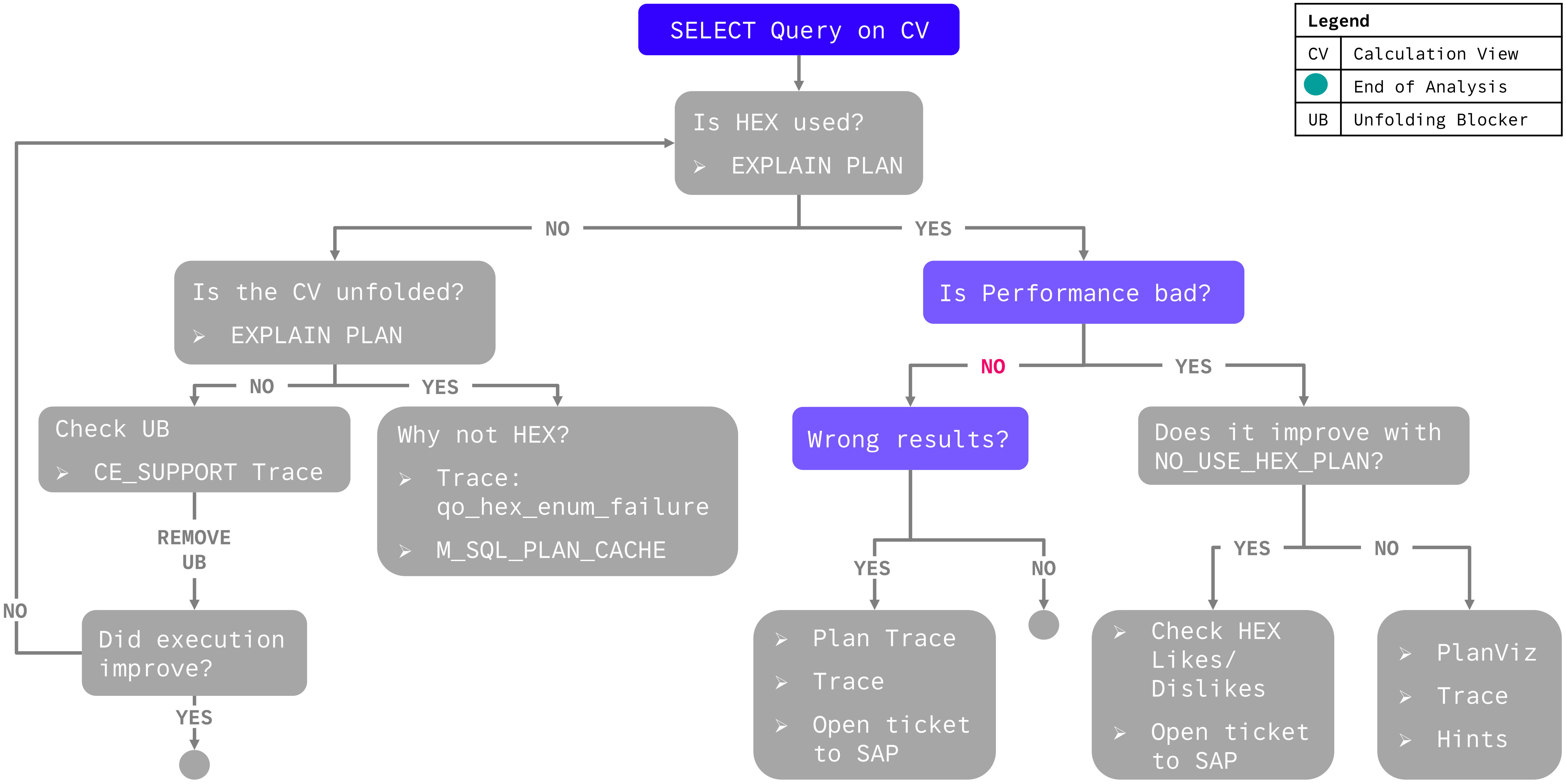


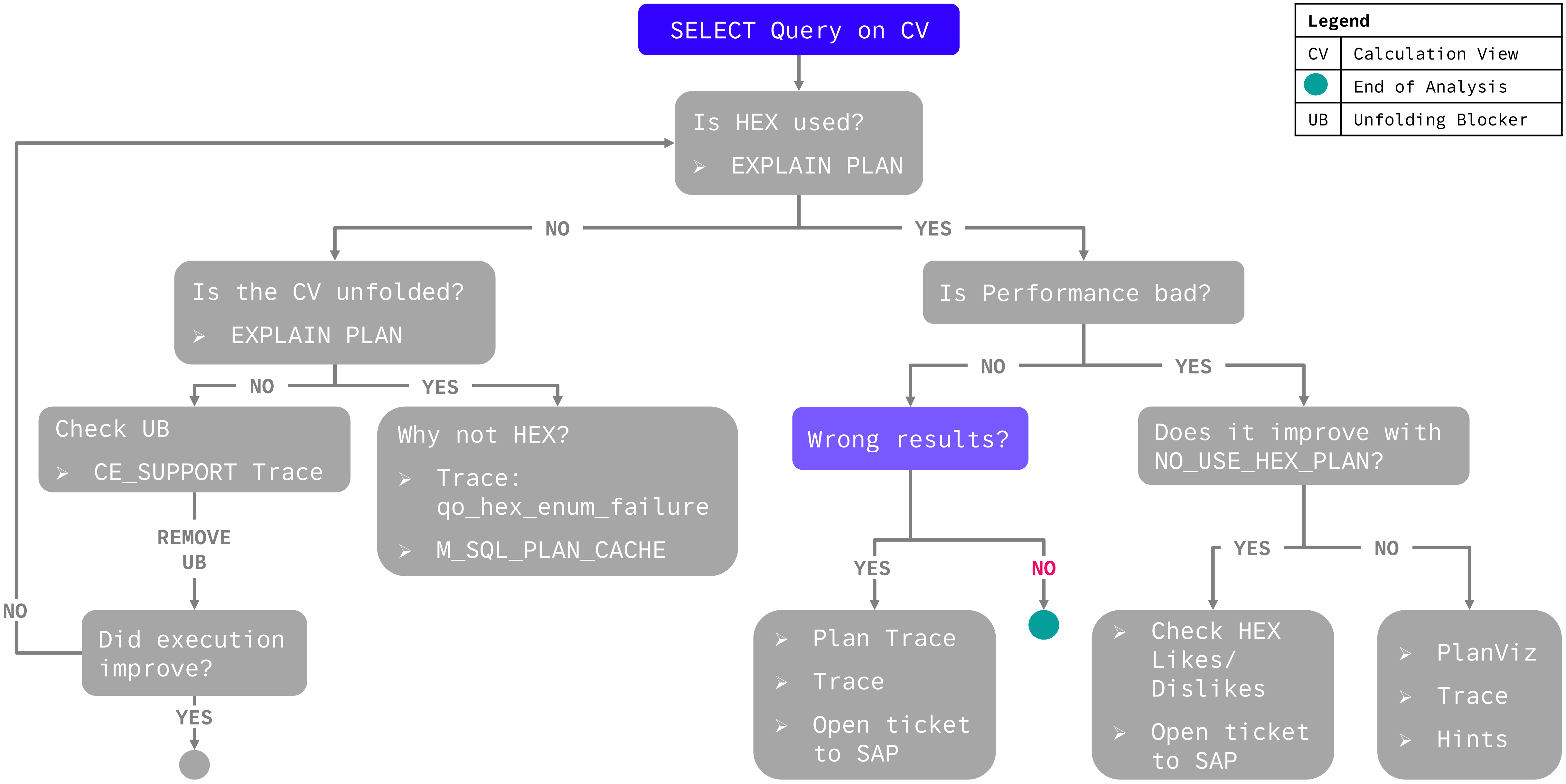
SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

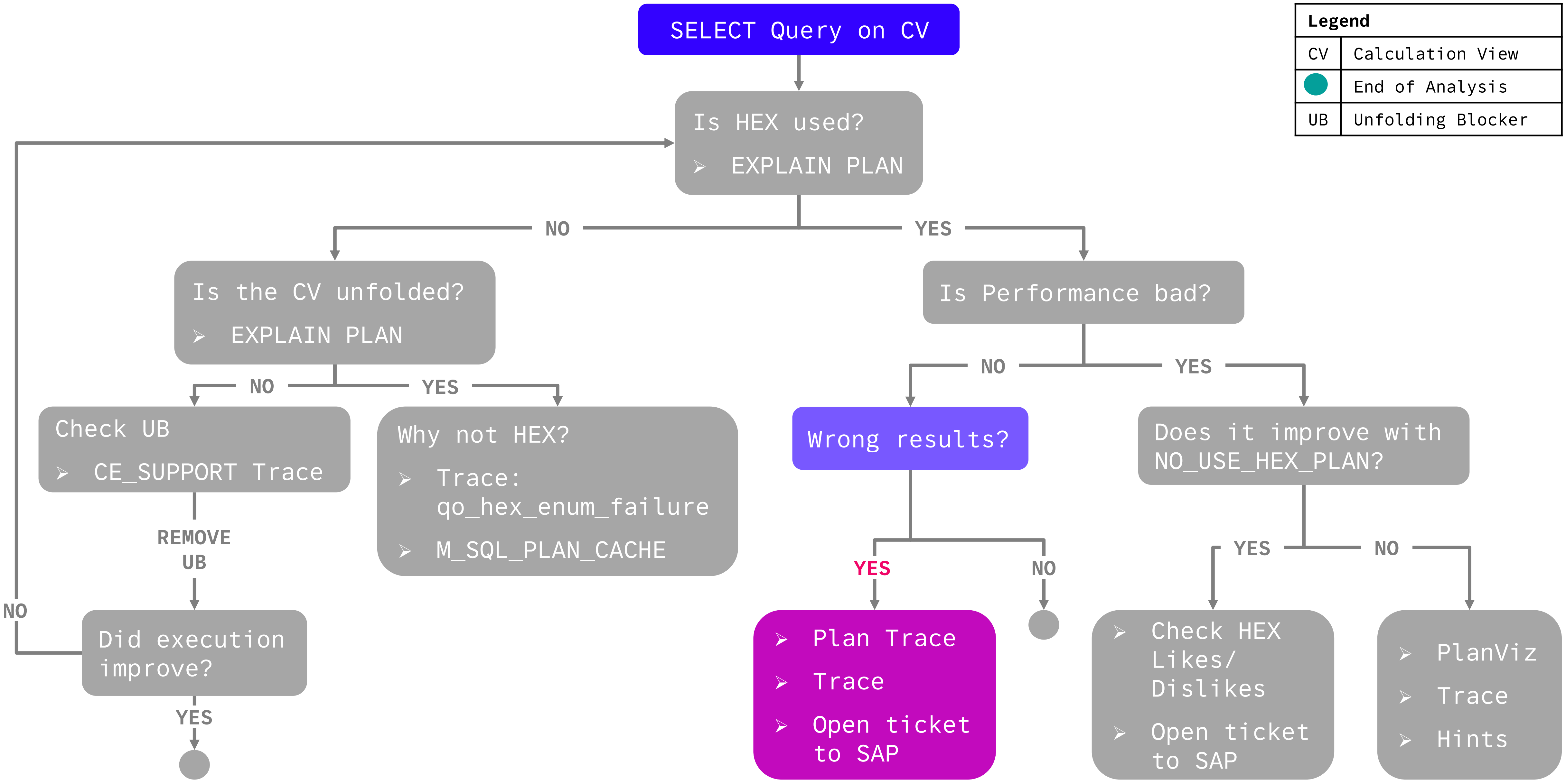
Is Performance Bad?





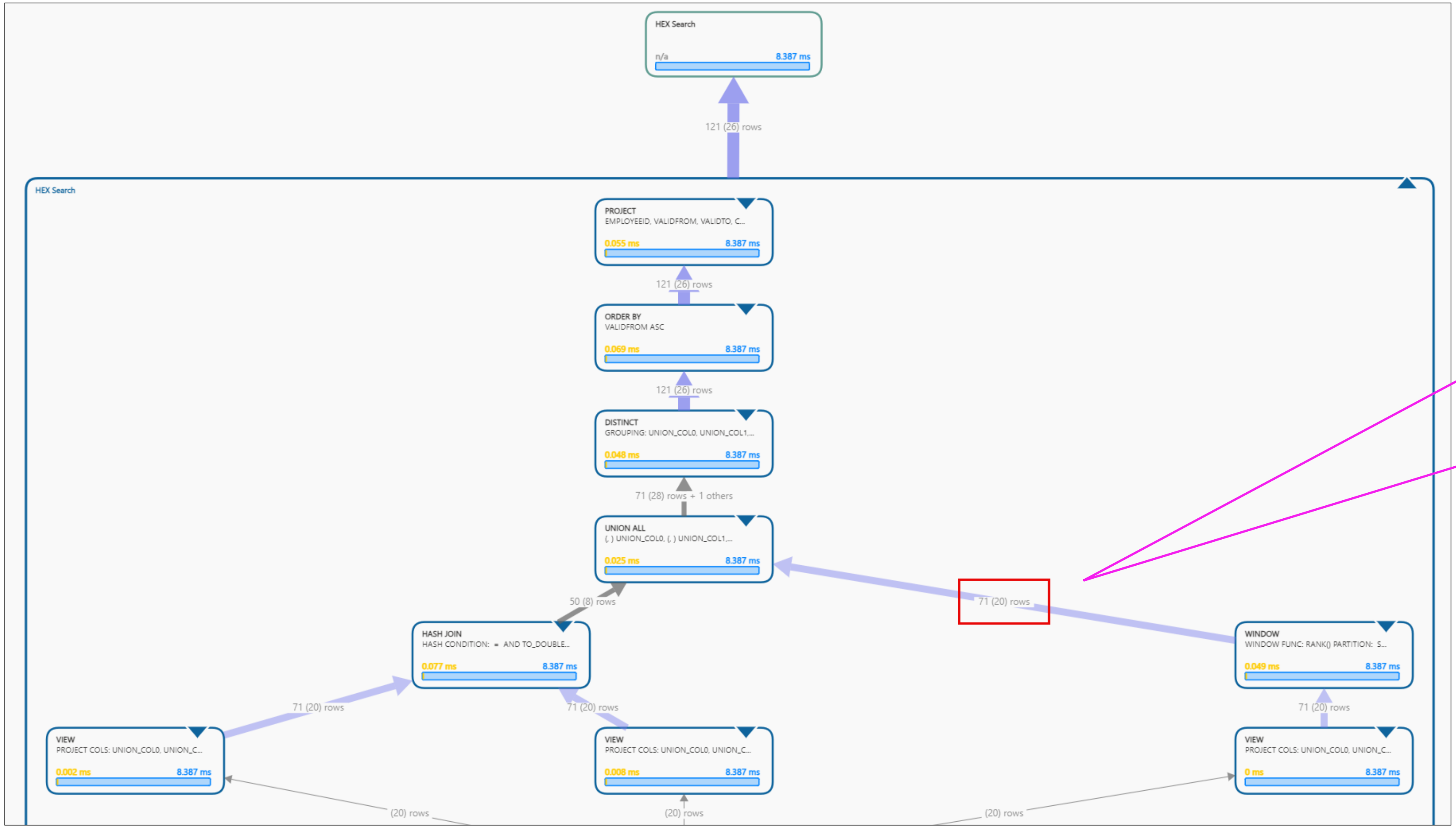

SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
Is Performance bad?
NO
YES
Check UB
➢ CE_SUPPORT Trace
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Wrong results?
Does it improve with NO_USE_HEX_PLAN?
REMOVE UB
Did execution improve?
YES
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

Wrong Results?

Plan Trace / PlanViz

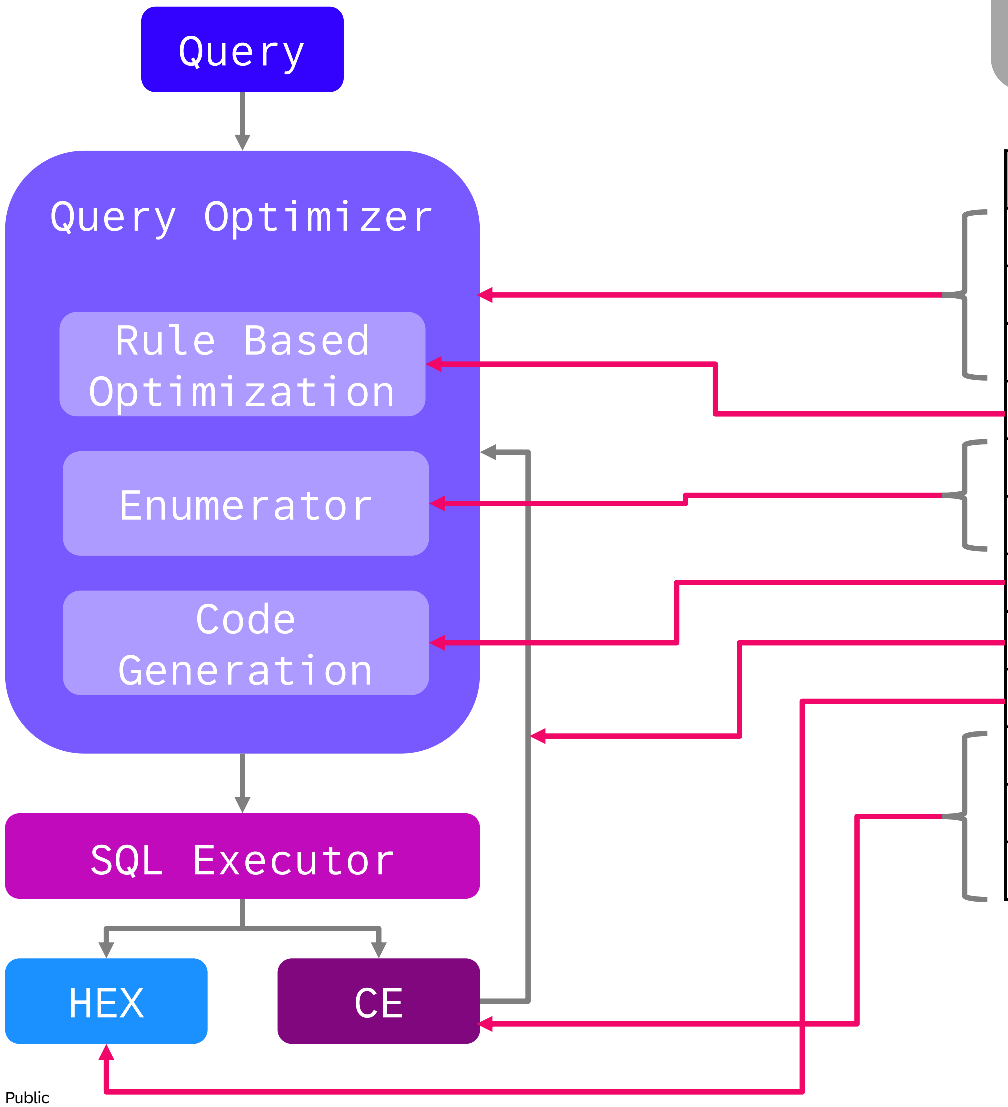


With knowledge of the view, or by comparing with a working version, we can find the node where there is an issue.

In this case, we see that **71** records are coming out of the Window Function, where **1** is expected

Wrong Results?

Tracers in the Context of HEX




```
ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM' ) SET
('trace', 'tracer') = 'level'
WITH RECONFIGURE;
```

Application	Tracer	Level
SQL Optimization Step Trace	SqlOptStep	debug
SQL Optimization Time Trace	SqlOpttime	debug
Cardinality Estimation Trace	SqlSizeEst	debug
Unfolding Trace	SqlOpt_ceqo	debug
Enum Helper Trace	plan_enum_helper	debug
HEX enumeration trace	qo_hex	debug
Codegen debug Trace	sqlopt_qe_gen	debug
Qo translation during ceqo	ceqo	debug
Hex Trace	hex	on / info
Calculation Engine Trace	calcengine	debug
Calculation Optimizer	ceoptimizer	debug
Calculation instantiate	ceinstantiate	debug

! No need to set **global.ini** in HANA Cloud

Wrong Results?

Trace Example for Calculation Engine (CE)

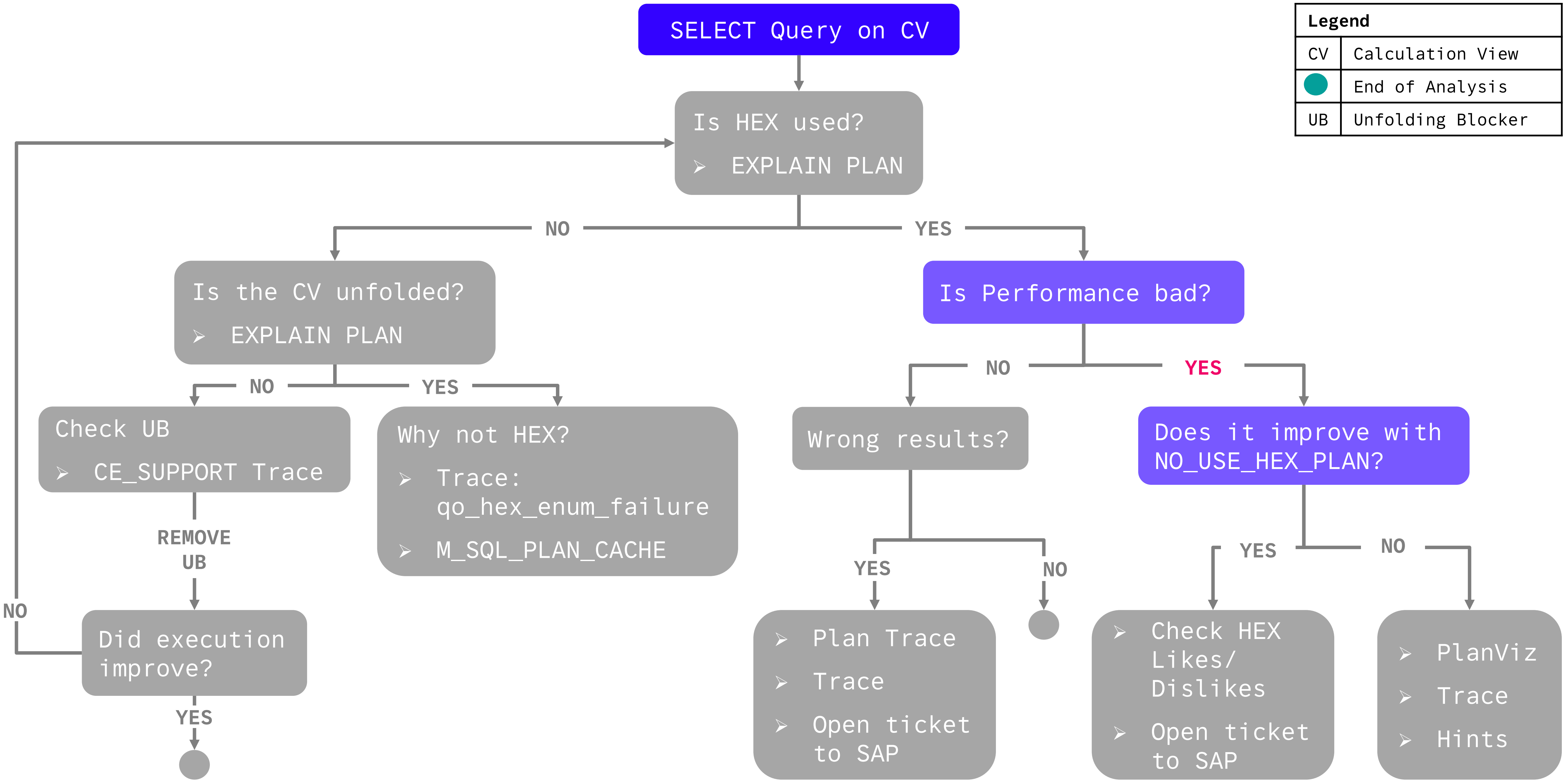
```
ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM') SET
('traceprofile_CE', 'calcengine') = 'debug',
('traceprofile_CE', 'ceinstantiate') = 'debug',
('traceprofile_CE', 'ceoptimizer') = 'debug',
('traceprofile_CE', 'ceoptimizerinfo') = 'info',
('traceprofile_CE', 'ceqo') = 'debug'
WITH RECONFIGURE;

-- <Query>

ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM') UNSET
('traceprofile_CE')
WITH RECONFIGURE;
```

Wrong Results?

Open Ticket to SAP

What to Include

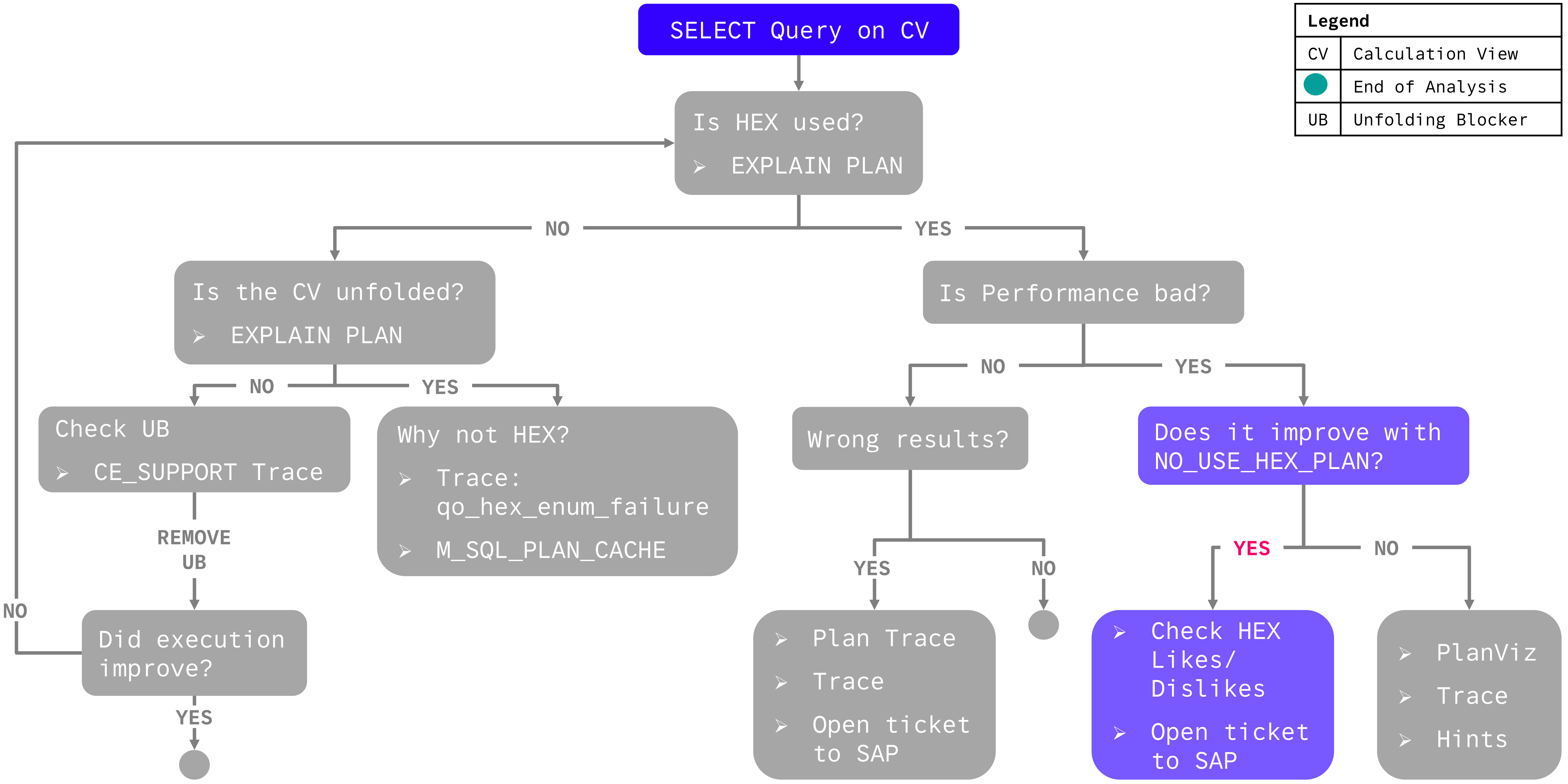
- Context information: system version, recent changes to; system, query, or data
- PlanViz file correct results + Hint NO_HEX_FUSE_JIT_CODE (if possible/available)
- PlanViz file wrong results + Hint NO_HEX_FUSE_JIT_CODE
- EXPLAIN PLAN correct results (if possible/available)
- EXPLAIN PLAN wrong results
- Trace with tracers:
  - hex = info
  - qo_hex = debug
  - qe_hex = info
  - sqloptstep = debug


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

Does it Improve with Hint NO_USE_HEX_PLAN?

```
SELECT *
FROM "HANA_TEC_CON_2025_HDI_DB01_1"."CV_BOOKS"
WITH HINT (NO_USE_HEX_PLAN);
```

Generally a good analysis step to confirm or eliminate the use of the HEX Engine as the root cause of the issue.


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

When Legacy Engines Perform Better than HEX

❌ **Dislikes (Performance Killers)**

Partial unfolding (e.g., nested CalcViews not unfolded)

Procedural SQLScript logic (FOR loops, cursors, etc.)

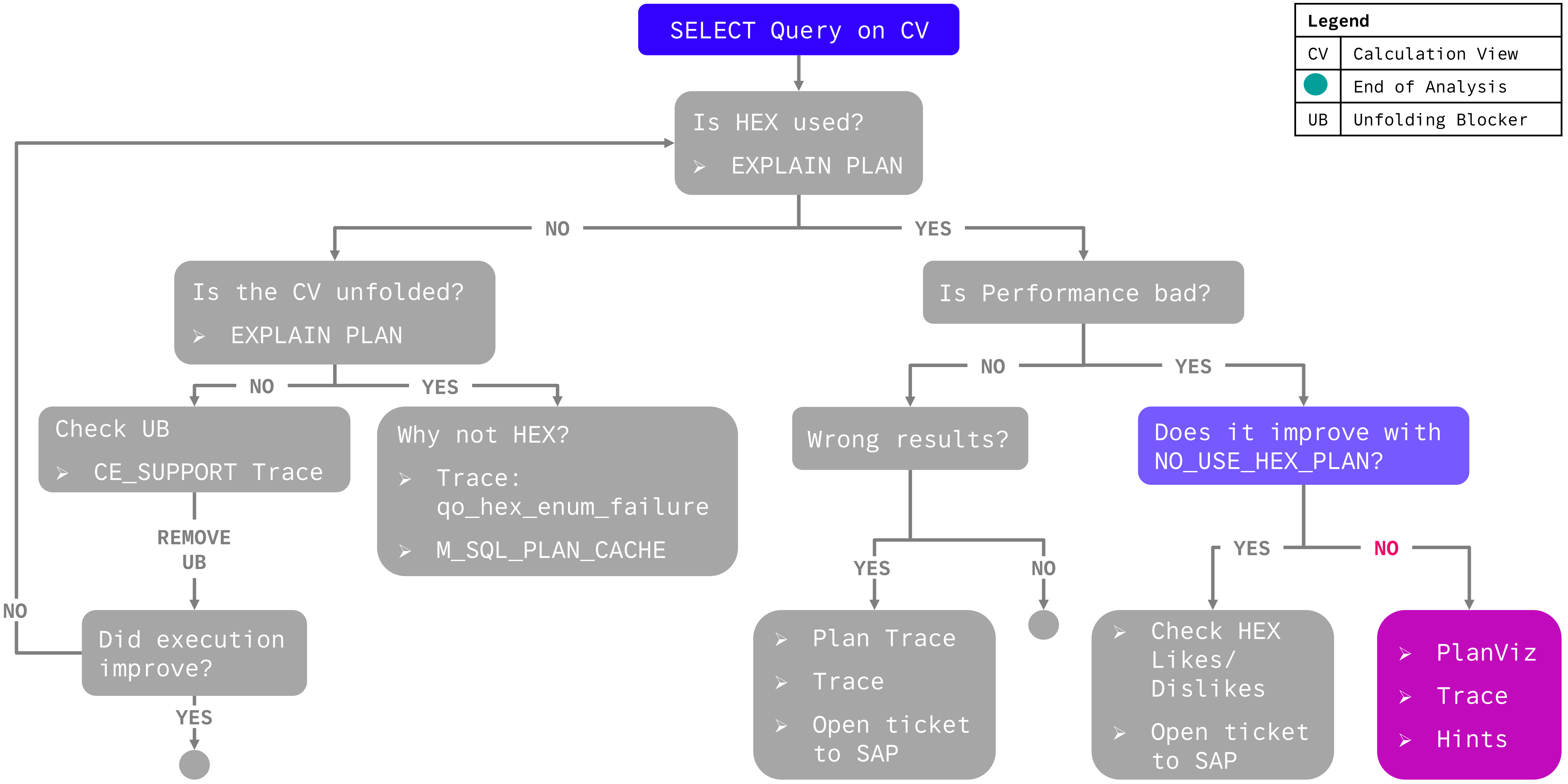
Skewed data (e.g., one customer has 99% of the records)

Rigid model logic that prevents rule-based optimizations

Could one of these conditions hold true?

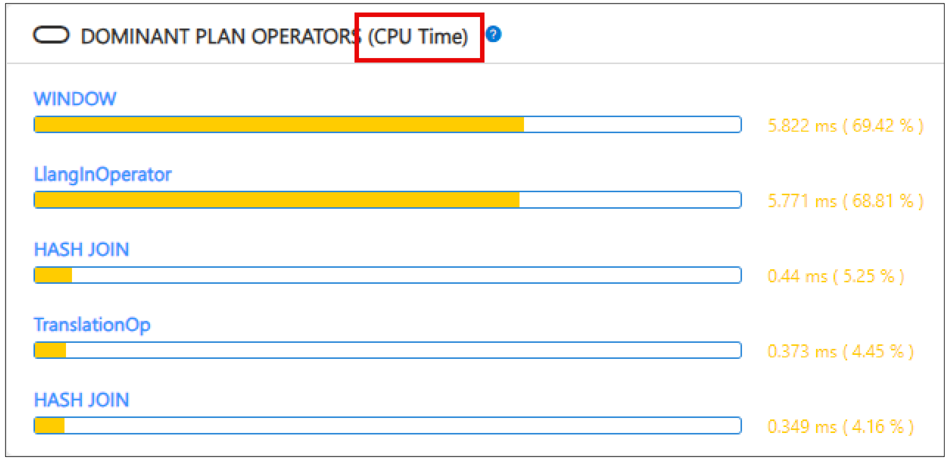
Options

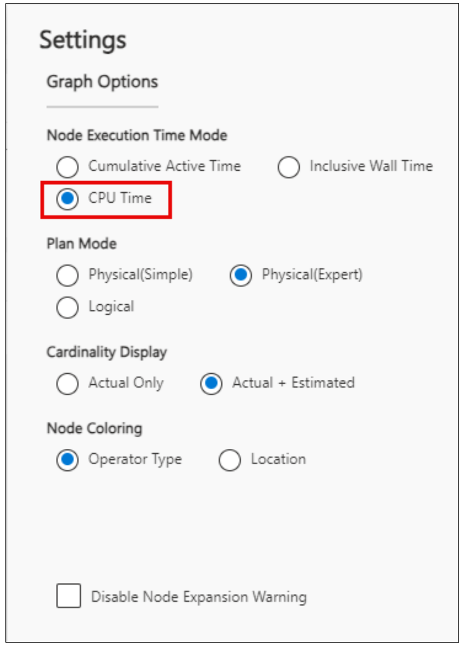
- Apply the hint NO_USE_HEX_PLAN to the CV + check with each upgrade if performance improves
- Using NO_USE_HEX_PLAN is not acceptable → Open a support ticket with SAP


SELECT Query on CV
Is HEX used?
➢ EXPLAIN PLAN
NO
YES
Is the CV unfolded?
➢ EXPLAIN PLAN
NO
YES
Check UB
➢ CE_SUPPORT Trace
REMOVE UB
Did execution improve?
NO
YES
Why not HEX?
➢ Trace: qo_hex_enum_failure
➢ M_SQL_PLAN_CACHE
Is Performance bad?
NO
YES
Wrong results?
YES
NO
➢ Plan Trace
➢ Trace
➢ Open ticket to SAP
Does it improve with NO_USE_HEX_PLAN?
YES
NO
➢ Check HEX Likes/ Dislikes
➢ Open ticket to SAP
➢ PlanViz
➢ Trace
➢ Hints
Legend
CV Calculation View
End of Analysis
UB Unfolding Blocker

When HEX is Slow

PlanViz / SQL Analyzer

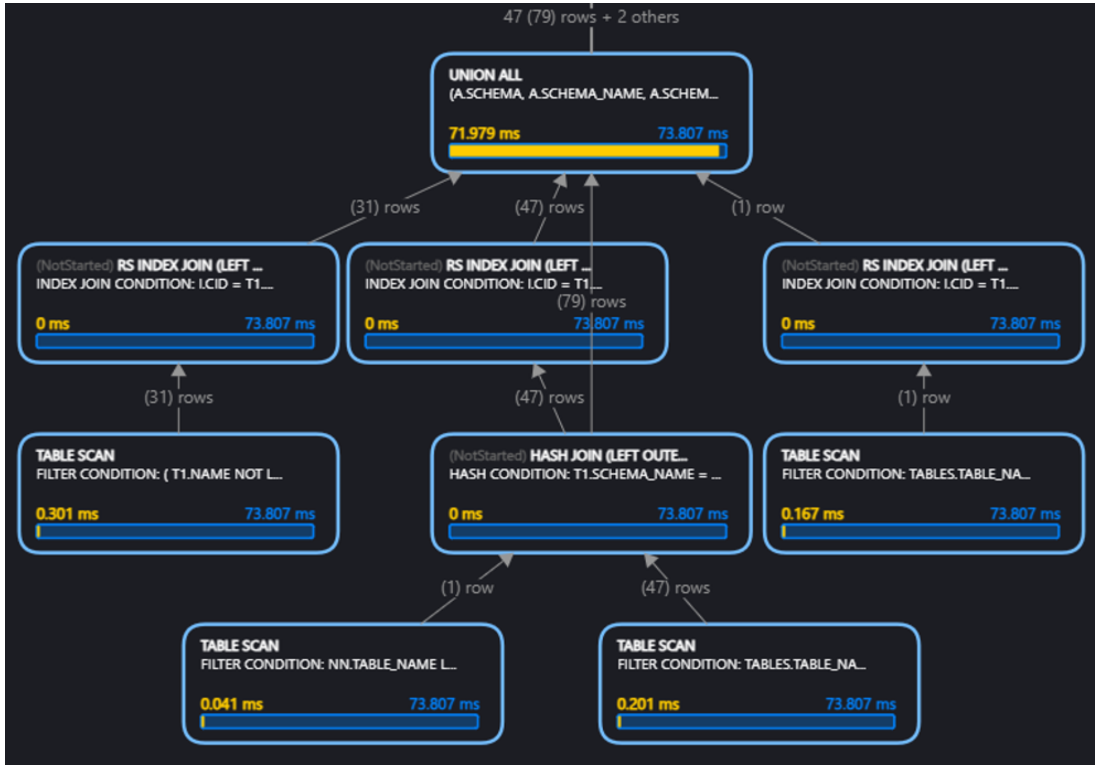




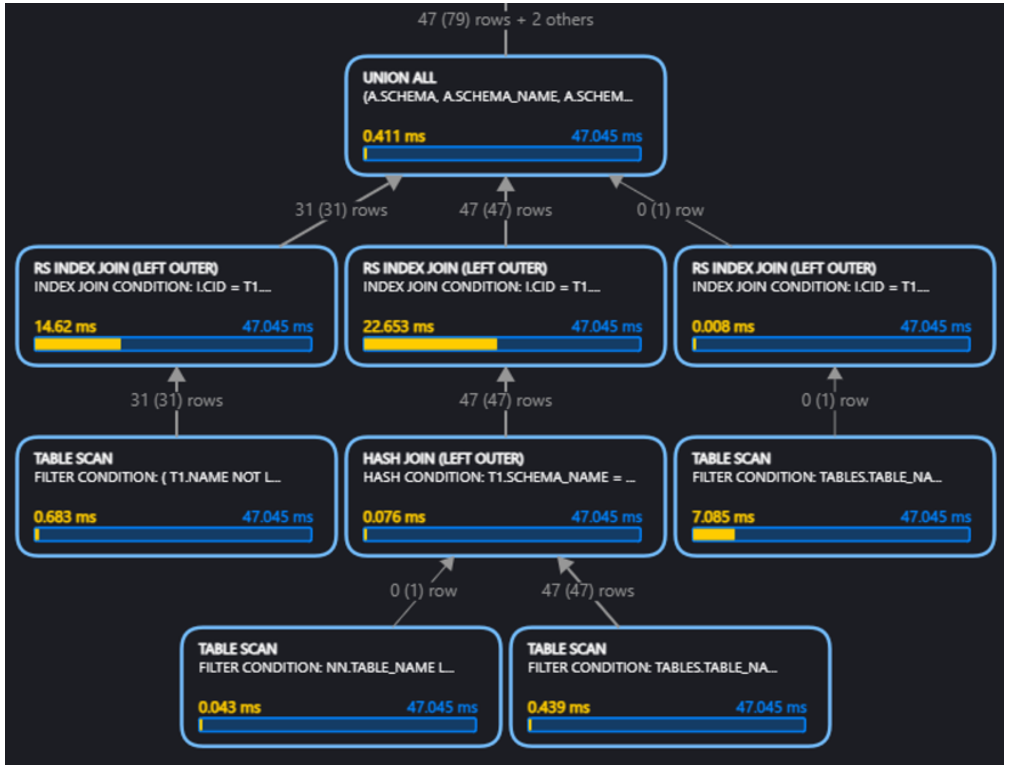
When HEX is Slow

PlanViz File

No Hint



HEX_NO_FUSE_JIT_CODE



When HEX is Slow

Trace: sqlopttime

```
--------------------------------------------------------------------------
* Parsing time:                                        0.263 ms
* Preprocessing time:                                  1.898 ms
* Checking time:                                       6.458 ms
* Related objects collection time:                     1.368 ms
* QP to QC conversion time:                            0.265 ms
* QC to QO conversion time:                            0.110 ms
* Query rewriting time:                            34355.665 ms
** Number of operators:                                   12
** Memory pool size:                                 503.012 MB
** Number of transformations:                              1
** Number of rollbacks:                                    0
* Cost-based optimization time:                       17.943 ms
** Local filter selectivity estimation time:          10.027 ms
** Join selectivity estimation time:                   0.000 ms
** ColJoinEstimator calls time:                        2.372 ms
** Number of JOIN_THRU_JOIN:                               8
** Number of JOIN_THRU_UNION_ALL:                          2
** Number of FILTER_THRU_JOIN:                             1
** Number of RS_PROJECT:                                   2
** Number of RS_TABLE_SCAN:                                7
** Number of CS_FILTER:                                    4
** Number of CS_JOIN:                                     24
** Number of CS_UNION_ALL:                                 4
** Number of CS_MATERIALIZE:                               3
** Number of CS_TABLE:                                     7
** Number of CS_EXPR_JOIN:                                24
** Number of COMPOSE_REMOTE_SEARCH:                        7
** Number of ESX_FILTER:                                   4
** Number of ESX_JOIN:                                    24
** Number of ESX_UNION_ALL:                                4
```

```
=== RewriteRule Time Analysis ===
[Pushdown Pre-aggregation to Target Table] Prepare: 0.00 ms, MainBody: 0.01 ms, Finalize: 0.00 ms, success: False
...
[View Unfolding                           ] Prepare: 0.00 ms, MainBody: 12525.95 ms, Finalize: 0.00 ms, success: True
[Collect dependent tables for cache       ] Prepare: 0.00 ms, MainBody: 16.67 ms, Finalize: 0.00 ms, success: False
[Evaluate args that needs to be const     ] Prepare: 0.00 ms, MainBody: 21.50 ms, Finalize: 0.00 ms, success: True
[Cleanup intermediate Project Cols        ] Prepare: 16.30 ms, MainBody: 91.28 ms, Finalize: 0.00 ms, success: True
[Remove Unnecessary Col                   ] Prepare: 15.71 ms, MainBody: 12986.90 ms, Finalize: 32.02 ms, success: True
[Join cardinality info removal with hint  ] Prepare: 0.00 ms, MainBody: 15.98 ms, Finalize: 0.00 ms, success: False
[Unnecessary Join Removal                 ] Prepare: 0.00 ms, MainBody: 1649.17 ms, Finalize: 426.34 ms, success: True
[Trex Key Marking                         ] Prepare: 0.00 ms, MainBody: 15.63 ms, Finalize: 0.00 ms, success: False
[Date/Time Formatting on Const            ] Prepare: 0.00 ms, MainBody: 15.89 ms, Finalize: 0.00 ms, success: False
[Predicate Normalization                  ] Prepare: 0.00 ms, MainBody: 2770.85 ms, Finalize: 32.19 ms, success: True
[Predicate Derivation                     ] Prepare: 0.00 ms, MainBody: 16.21 ms, Finalize: 32.31 ms, success: True
[Disjuntive Filter into Union All         ] Prepare: 0.00 ms, MainBody: 482.04 ms, Finalize: 48.11 ms, success: True
...
* Code generation time:                            8.829 ms
* Total query compilation time:                34400.526 ms
* partition calc api call time                     0.000 ms
** partition calc api call count                       0
* partition finalize api call time                 0.000 ms
** partition finalize api call count                   0
--------------------------------------------------------------------
```

When HEX is Slow

Trace Example for HEX

```
ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM') SET
('traceprofile_HEX', 'sql_user') = '<user>',
('traceprofile_HEX', 'hex') = 'info',
('traceprofile_HEX', 'qe_hex') = 'info',
('traceprofile_HEX', 'qo_hex') = 'debug',
('traceprofile_HEX', 'sqloptstep') = 'debug'
WITH RECONFIGURE;

-- <Query> WITH HINT (IGNORE_PLAN_CACHE);

ALTER SYSTEM ALTER CONFIGURATION ('indexserver.ini', 'SYSTEM') UNSET
('traceprofile_HEX')
WITH RECONFIGURE;
```

When HEX is Slow

Using Hints

Using the previous analysis, candidate hints can be found.

	OPERATOR_NAME	OPERATOR_DETAILS	OPERATOR_PROPERTIES	EXECUTION_ENGINE	DATABASE_NAME	SCHEMA_NAME
1	PROJECT	BIMC_VARIABLE_VIEW_INT.CATALOG_NAME, BIMC_VARIABLE_VIEW_INT.CUBE_NAME, BIM...	ATTACHED HINT LIST (USE_HEX_PLAN, OPTIMIZATION_LEVEL(RULE_BASED)), PHYSICAL_ENUM_BY: HEX_PROJECT	HEX	?	?
2	ORDER BY	BIMC_VARIABLE_VIEW_INT.ORDER ASC	PHYSICAL_ENUM_BY: HEX_ORDER_BY	HEX	?	?
3	HASH JOIN	HASH CONDITION: VIEWS.SCHEMA_NAME = C.SCHEMA_NAME AND VIEWS.VIEW_NAME ...	PHYSICAL_ENUM_BY: HEX_HASH_JOIN	HEX	?	?
4	UNION ALL	(CS_JOIN_INDEXES_.SCHEMA_NAME, CS_CALC_INDEXES_.SCHEMA_NAME, CS_HIERARCH...	PHYSICAL_ENUM_BY: HEX_UNION_ALL	HEX	?	?
5	TABLE SCAN		PHYSICAL_ENUM_BY: HEX_RS_TABLE_SCAN	HEX		SYS
6	TABLE SCAN	FILTER CONDITION: CS_CALC_INDEXES_.VIEW_NAME NOT LIKE '_SYS%' ESCAPE '\'	PHYSICAL_ENUM_BY: HEX_RS_TABLE_SCAN	HEX		SYS
7	TABLE SCAN		PHYSICAL_ENUM_BY: HEX_RS_TABLE_SCAN	HEX		SYS
8	HASH JOIN	HASH CONDITION: BIMC_VARIABLE_VIEW_INT.CATALOG_NAME = BIMC_AUTH_ONLY_CU...	TRANSLATION TABLE (LEFT): BIMC_VARIABLE_VIEW_INT.CATALOG_NAME = BIMC_AUTH_ONLY_CUBES.CATALOG_NAME, BIMC_...	HEX	?	?
9	INDEX JOIN (LEFT OUTER)	INDEX JOIN CONDITION: __cs_concat__(BIMC_VARIABLE_VIEW_INT.SCHEMA_NAME, BIMC...	PHYSICAL_ENUM_BY: HEX_INDEX_JOIN	HEX		_SYS_BI
10	INDEX JOIN (LEFT OUTER)	INDEX JOIN CONDITION: __cs_concat__(BIMC_VARIABLE_VIEW_INT.SCHEMA_NAME, BIMC...	PHYSICAL_ENUM_BY: HEX_INDEX_JOIN	HEX		_SYS_BI
11	INDEX JOIN	INDEX JOIN CONDITION: BIMC_VARIABLE_ASSIGNMENT.$trexexternalkey$ BETWEEN __cs_...	PHYSICAL_ENUM_BY: HEX_INDEX_JOIN	HEX		_SYS_BI
12	TABLE SCAN	FILTER CONDITION: BIMC_VARIABLE_VIEW_INT.IS_INPUT_ENABLED = 1 (DETAIL: ([SCAN] BI...	PHYSICAL_ENUM_BY: HEX_TABLE_SCAN	HEX		_SYS_BI
13	TABLE SCAN		PHYSICAL_ENUM_BY: HEX_TABLE_SCAN	HEX		_SYS_BI

Example: NO_HEX_INDEX_JOIN

Analyze Current schema: Connected to:

```sql
SELECT * FROM HINTS WHERE HINT_NAME LIKE '%HEX%';
```

Result x Messages x History

Rows (77) SQL

	HINT_NAME	DESCRIPTION
1	HEX_LIMIT_THRU_OUTER_JOIN	Prefer HEX limit thru outer join
2	NO_HEX_LIMIT_THRU_OUTER_JOIN	Avoid HEX limit thru outer join
3	USE_HEX_PLAN	Prefer new column engine
4	NO_USE_HEX_PLAN	Avoid new column engine
5	HEX_NESTED_LOOP_JOIN	Prefer HEX nested loop join
6	NO_HEX_NESTED_LOOP_JOIN	Avoid HEX nested loop join
7	HEX_INDEX_JOIN	Prefer HEX index join
8	NO_HEX_INDEX_JOIN	Avoid HEX index join
9	HEX_TABLE_SCAN_SEMI_JOIN	Prefer HEX table scan semi join

Analyze

```sql
SELECT * FROM STATEMENT_HINTS;
```

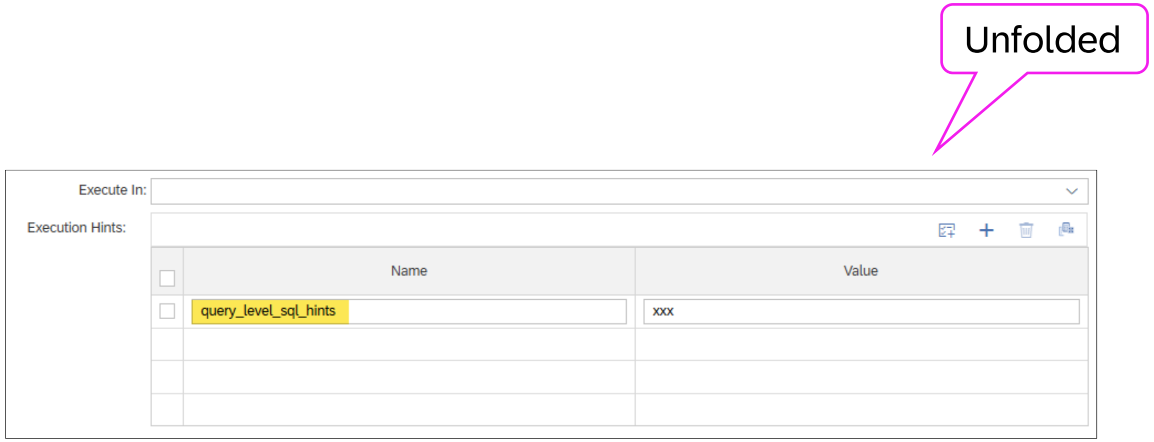
Result x Messages x History

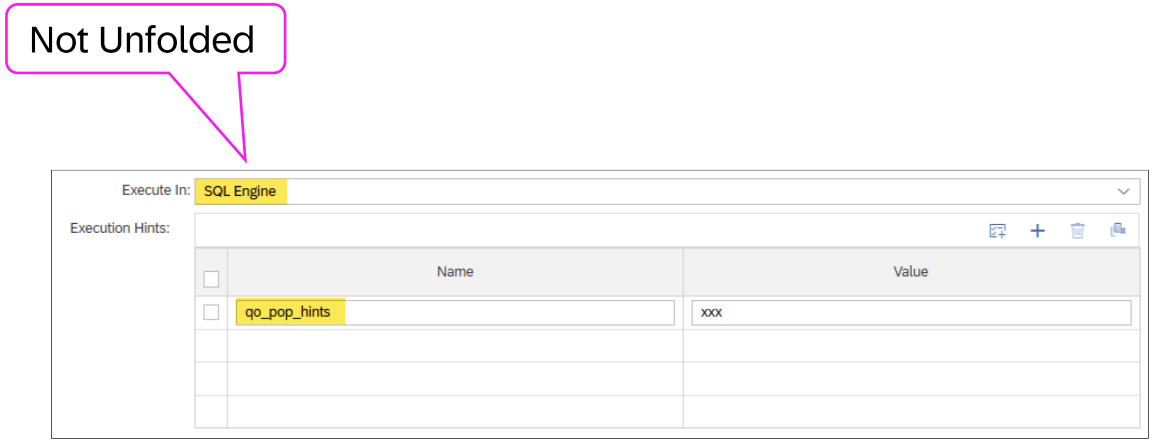
Rows (1)

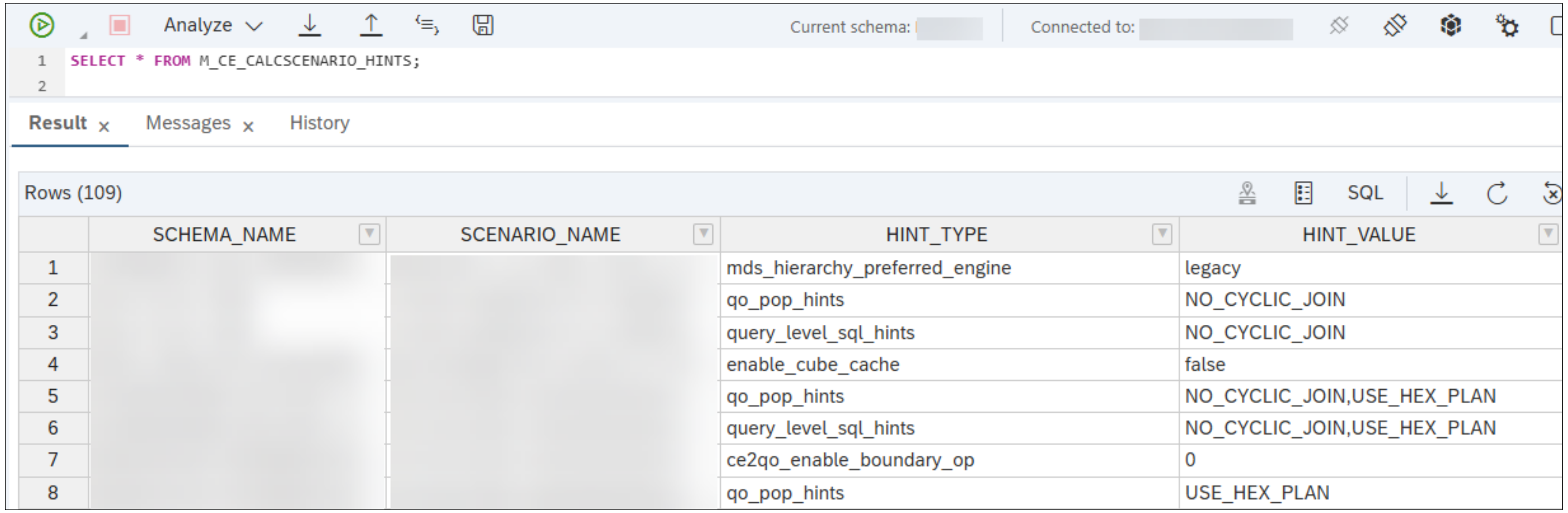
	STATEMENT_STRING	STATEMENT_HASH	HINT_STRING	SYSTEM_HINT_STRING	COMMENTS	IS_ENABLED
1	SELECT * FROM "HANA_TEC_CON_2025_HDI_DB01_1"."CV_BOOKS"	4b79809b430937836c46a79c4abbfdbd	NO_HEX_INDEX_JOIN			TRUE

When HEX is Slow

Hints on Calculation Views








```sql
SELECT * FROM M_CE_CALCSCENARIO_HINTS;
```

	SCHEMA_NAME	SCENARIO_NAME	HINT_TYPE	HINT_VALUE
1			mds_hierarchy_preferred_engine	legacy
2			qo_pop_hints	NO_CYCLIC_JOIN
3			query_level_sql_hints	NO_CYCLIC_JOIN
4			enable_cube_cache	false
5			qo_pop_hints	NO_CYCLIC_JOIN,USE_HEX_PLAN
6			query_level_sql_hints	NO_CYCLIC_JOIN,USE_HEX_PLAN
7			ce2qo_enable_boundary_op	0
8			qo_pop_hints	USE_HEX_PLAN

Thank you.

Contact information:

Mohamed Ali Chriki
mohamed.ali.chriki@sap.com

Manuel Mayr
m.mayr@sap.com

Elisabeth Badino
elisabeth.badino@sap.com

SAP Bring out your best.

Appendix - Resources

Sources
CALCULATION ENGINE

[Checking Whether a Query is Unfolded]

[Influencing Whether a Query is Unfolded]

[Understand Why a Query Is Not Unfolded]

SAP Notes

2594739 - HANA Calculation Engine Support Mode Tracing

Open SAP Course

[Working with Calculation Views in SAP HANA Cloud]

Sources

SQL Optimizer & HEX

Open SAP Course

[A First Step Towards SAP HANA Query Optimization]

Sources

TROUBLESHOOTING

SAP HANA Troubleshooting and Performance Analysis Guide

SAP HANA Cloud, SAP HANA Database Performance Guide for Developers

2000002 - FAQ: SAP HANA SQL Optimization

Sources

EXPLAIN PLAN

SAP HANA Cloud, SAP HANA Database SQL Reference Guide → EXPLAIN PLAN Statement (Data Manipulation)

Sources

SQL ANALYZER

SQL Analyzer Tool for SAP HANA Guide

Sources
TRACING

SAP Notes

2909779 - How to Collect Frequently Required Debug Info for Analyzing HANA Issues

2380176 - FAQ: SAP HANA Database Trace

Sources
HINTS

HINT Details

SAP Notes

2142945 - FAQ: SAP HANA Hints

2400006 - FAQ: SAP HANA Statement Hints

2700051 - Delivery of SAP HANA Statement Hints